

CHAPTER

13

Properties of Triangle

Section-A

JEE Advanced/ IIT-JEE

A

Fill in the Blanks

1. In a $\triangle ABC$, $\angle A = 90^\circ$ and AD is an altitude. Complete the relation

$$\frac{BD}{BA} = \frac{AB}{(\dots)} \quad (1980)$$

2. ABC is a triangle, P is a point on AB , and Q is point on AC such that $\angle AQP = \angle ABC$. Complete the relation

$$\frac{\text{area of } \triangle APQ}{\text{area of } \triangle ABC} = \frac{(\dots)}{AC^2} \quad (1980)$$

3. ABC is a triangle with $\angle B$ greater than $\angle C$. D and E are points on BC such that AD is perpendicular to BC and AE is the bisector of angle A . Complete the relation (1980)

$$\angle DAE = \frac{1}{2}[(\dots) - \angle C]$$

4. The set of all real numbers a such that $a^2 + 2a$, $2a + 3$ and $a^2 + 3a + 8$ are the sides of a triangle is (1985 - 2 Marks)

5. In a triangle ABC , if $\cot A$, $\cot B$, $\cot C$ are in A.P., then a^2 , b^2 , c^2 , are in progression. (1985 - 2 Marks)

6. A polygon of nine sides, each of length 2, is inscribed in a circle. The radius of the circle is (1987 - 2 Marks)

7. If the angles of a triangle are 30° and 45° and the included side is $(\sqrt{3} + 1)$ cms, then the area of the triangle is (1988 - 2 Marks)

8. If in a triangle ABC , $\frac{2 \cos A}{a} + \frac{\cos B}{b} + \frac{2 \cos C}{c} = \frac{a}{bc} + \frac{b}{ca}$, then the value of the angle A is degrees. (1993 - 2 Marks)

9. In a triangle ABC , AD is the altitude from A . Given $b > c$, $\angle C = 23^\circ$ and $AD = \frac{abc}{b^2 - c^2}$ then $\angle B = \dots\dots\dots$ (1994 - 2 Marks)

10. A circle is inscribed in an equilateral triangle of side a . The area of any square inscribed in this circle is (1994 - 2 Marks)

11. In a triangle ABC , $a : b : c = 4 : 5 : 6$. The ratio of the radius of the circumcircle to that of the incircle is (1996 - 1 Mark)

C

MCQs with One Correct Answer

1. If the bisector of the angle P of a triangle PQR meets QR in S , then (1979)

- (a) $QS = SR$ (b) $QS : SR = PR : PQ$
(c) $QS : SR = PQ : PR$ (d) None of these

2. From the top of a light-house 60 metres high with its base at the sea-level, the angle of depression of a boat is 15° . The distance of the boat from the foot of the light house is (1983 - 1 Mark)

- (a) $\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right)$ 60 metres (b) $\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)$ 60 metres

- (c) $\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right)^2$ metres (d) none of these

3. In a triangle ABC , angle A is greater than angle B . If the measures of angles A and B satisfy the equation $3 \sin x - 4 \sin^3 x - k = 0$, $0 < k < 1$, then the measure of angle C is (1990 - 2 Marks)

- (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$ (c) $\frac{2\pi}{3}$ (d) $\frac{5\pi}{6}$

4. If the lengths of the sides of triangle are 3, 5, 7 then the largest angle of the triangle is (1994)

- (a) $\frac{\pi}{2}$ (b) $\frac{5\pi}{6}$ (c) $\frac{2\pi}{3}$ (d) $\frac{3\pi}{4}$

5. In a triangle ABC , $\angle B = \frac{\pi}{3}$ and $\angle C = \frac{\pi}{4}$. Let D divide BC internally in the ratio 1 : 3 then $\frac{\sin \angle BAD}{\sin \angle CAD}$ is equal to (1995S)

- (a) $\frac{1}{\sqrt{6}}$ (b) $\frac{1}{3}$ (c) $\frac{1}{\sqrt{3}}$ (d) $\sqrt{\frac{2}{3}}$

6. In a triangle ABC , $2ac \sin \frac{1}{2}(A - B + C) =$ (2000S)

- (a) $a^2 + b^2 - c^2$ (b) $c^2 + a^2 - b^2$
(c) $b^2 - c^2 - a^2$ (d) $c^2 - a^2 - b^2$

7. In a triangle ABC , let $\angle C = \frac{\pi}{2}$. If r is the inradius and R is the circumradius of the triangle, then $2(r+R)$ is equal to
(2000S)

(a) $a+b$ (b) $b+c$ (c) $c+a$ (d) $a+b+c$

8. A pole stands vertically inside a triangular park $\triangle ABC$. If the angle of elevation of the top of the pole from each corner of the park is same, then in $\triangle ABC$ the foot of the pole is at the
(2000S)

(a) centroid (b) circumcentre
(c) incentre (d) orthocentre

9. A man from the top of a 100 metres high tower sees a car moving towards the tower at an angle of depression of 30° . After some time, the angle of depression becomes 60° . The distance (in metres) travelled by the car during this time is

(a) $100\sqrt{3}$ (b) $200\sqrt{3}/3$ (2001S)
(c) $100\sqrt{3}/3$ (d) $200\sqrt{3}$

10. Which of the following pieces of data does NOT uniquely determine an acute-angled triangle ABC (R being the radius of the circumcircle)?
(2002S)

(a) $a, \sin A, \sin B$ (b) a, b, c
(c) $a, \sin B, R$ (d) $a, \sin A, R$

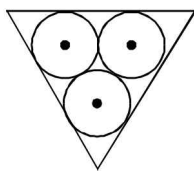
11. If the angles of a triangle are in the ratio $4:1:1$, then the ratio of the longest side to the perimeter is
(2003S)

(a) $\sqrt{3}:(2+\sqrt{3})$ (b) $1:6$
(c) $1:2+\sqrt{3}$ (d) $2:3$

12. The sides of a triangle are in the ratio $1:\sqrt{3}:2$, then the angles of the triangle are in the ratio
(2004S)

(a) $1:3:5$ (b) $2:3:4$ (c) $3:2:1$ (d) $1:2:3$

13. In an equilateral triangle, 3 coins of radii 1 unit each are kept so that they touch each other and also the sides of the triangle. Area of the triangle is
(2005S)



(a) $4+2\sqrt{3}$ (b) $6+4\sqrt{3}$
(c) $12+\frac{7\sqrt{3}}{4}$ (d) $3+\frac{7\sqrt{3}}{4}$

14. In a triangle ABC , a, b, c are the lengths of its sides and A, B, C are the angles of triangle ABC . The correct relation is given by
(2005S)

(a) $(b-c)\sin\left(\frac{B-C}{2}\right) = a\cos\frac{A}{2}$
(b) $(b-c)\cos\left(\frac{A}{2}\right) = a\sin\frac{B-C}{2}$

(c) $(b+c)\sin\left(\frac{B+C}{2}\right) = a\cos\frac{A}{2}$

(d) $(b-c)\cos\left(\frac{A}{2}\right) = 2a\sin\frac{B+C}{2}$

15. One angle of an isosceles $\triangle$ is 120° and radius of its incircle is $\sqrt{3}$. Then the area of the triangle in sq. units is
(2006 - 3M, -1)

(a) $7+12\sqrt{3}$ (b) $12-7\sqrt{3}$
(c) $12+7\sqrt{3}$ (d) 4π

16. Let $ABCD$ be a quadrilateral with area 18, with side AB parallel to the side CD and $2AB = CD$. Let AD be perpendicular to AB and CD . If a circle is drawn inside the quadrilateral $ABCD$ touching all the sides, then its radius is
(2007 - 3 marks)

(a) 3 (b) 2 (c) $\frac{3}{2}$ (d) 1

17. If the angles A, B and C of a triangle are in an arithmetic progression and if a, b and c denote the lengths of the sides opposite to A, B and C respectively, then the value of the expression $\frac{a}{c}\sin 2C + \frac{c}{a}\sin 2A$ is
(2010)

(a) $\frac{1}{2}$ (b) $\frac{\sqrt{3}}{2}$ (c) 1 (d) $\sqrt{3}$

18. Let PQR be a triangle of area Δ with $a=2$, $b=\frac{7}{2}$ and $c=\frac{5}{2}$; where a, b , and c are the lengths of the sides of the triangle opposite to the angles at P, Q and R respectively. Then $\frac{2\sin P - \sin 2P}{2\sin P + \sin 2P}$ equals.
(2012)

(a) $\frac{3}{4\Delta}$ (b) $\frac{45}{4\Delta}$ (c) $\left(\frac{3}{4\Delta}\right)^2$ (d) $\left(\frac{45}{4\Delta}\right)^2$

19. In a triangle the sum of two sides is x and the product of the same sides is y . If $x^2 - c^2 = y$, where c is the third side of the triangle, then the ratio of the in radius to the circum-radius of the triangle is
(JEE Adv. 2014)

(a) $\frac{3y}{2x(x+c)}$ (b) $\frac{3y}{2c(x+c)}$
(c) $\frac{3y}{4x(x+c)}$ (d) $\frac{3y}{4c(x+c)}$

D MCQs with One or More than One Correct

1. There exists a triangle ABC satisfying the conditions
(1986 - 2 Marks)
- (a) $b \sin A = a, A < \pi/2$
(b) $b \sin A > a, A > \pi/2$
(c) $b \sin A > a, A < \pi/2$
(d) $b \sin A < a, A < \pi/2, b > a$
(e) $b \sin A < a, A > \pi/2, b = a$

Properties of Triangle

2. In a triangle, the lengths of the two larger sides are 10 and 9, respectively. If the angles are in A.P. Then the length of the third side can be (1987 - 2 Marks)
- (a) $5 - \sqrt{6}$ (b) $3\sqrt{3}$
 (c) 5 (d) $5 + \sqrt{6}$ (e) none
3. If in a triangle PQR , $\sin P$, $\sin Q$, $\sin R$ are in A.P., then (1998 - 2 Marks)
- (a) the altitudes are in A.P. (b) the altitudes are in H.P.
 (c) the medians are in G.P. (d) the medians are in A.P.
4. Let $A_0A_1A_2A_3A_4A_5$ be a regular hexagon inscribed in a circle of unit radius. Then the product of the lengths of the line segments A_0A_1 , A_0A_2 and A_0A_4 is (1998 - 2 Marks)
- (a) $\frac{3}{4}$ (b) $3\sqrt{3}$ (c) 3 (d) $\frac{3\sqrt{3}}{2}$
5. In $\triangle ABC$, internal angle bisector of $\angle A$ meets side BC in D . $DE \perp AD$ meets AC in E and AB in F . Then (2006 - 5M, -1)
- (a) AE is HM of b & c (b) $AD = \frac{2bc}{b+c} \cos \frac{A}{2}$
 (c) $EF = \frac{4bc}{b+c} \sin \frac{A}{2}$ (d) $\triangle AEF$ is isosceles
6. Let ABC be a triangle such that $\angle ACB = \frac{\pi}{6}$ and let a , b and c denote the lengths of the sides opposite to A , B and C respectively. The value(s) of x for which $a = x^2 + x + 1$, $b = x^2 - 1$ and $c = 2x + 1$ is (are) (2010)
- (a) $-(2 + \sqrt{3})$ (b) $1 + \sqrt{3}$
 (c) $2 + \sqrt{3}$ (d) $4\sqrt{3}$
7. In a triangle PQR , P is the largest angle and $\cos P = \frac{1}{3}$. Further the incircle of the triangle touches the sides PQ , QR and RP at N , L and M respectively, such that the lengths of PN , QL and RM are consecutive even integers. Then possible length(s) of the side(s) of the triangle is (are) (JEE Adv. 2013)
- (a) 16 (b) 18 (c) 24 (d) 22
8. In a triangle XYZ , let x , y , z be the lengths of sides opposite to the angles X , Y , Z , respectively, and $2s = x + y + z$. If $\frac{s-x}{4} = \frac{s-y}{3} = \frac{s-z}{2}$ and area of incircle of the triangle XYZ is $\frac{8\pi}{3}$, then (JEE Adv. 2016)
- (a) area of the triangle XYZ is $6\sqrt{6}$
 (b) the radius of circumcircle of the triangle XYZ is $\frac{35}{6}\sqrt{6}$
 (c) $\sin \frac{X}{2} \sin \frac{Y}{2} \sin \frac{Z}{2} = \frac{4}{35}$
 (d) $\sin^2 \left(\frac{X+Y}{2} \right) = \frac{3}{5}$

E

Subjective Problems

1. A triangle ABC has sides $AB = AC = 5$ cm and $BC = 6$ cm. Triangle $A'B'C'$ is the reflection of the triangle ABC in a line parallel to AB placed at a distance 2 cm from AB , outside the triangle ABC . Triangle $A''B''C''$ is the reflection of the triangle $A'B'C'$ in a line parallel to $B'C'$ placed at a distance of 2 cm from $B'C'$ outside the triangle $A'B'C'$. Find the distance between A and A'' . (1978)
2. (a) If a circle is inscribed in a right angled triangle ABC with the right angle at B , show that the diameter of the circle is equal to $AB + BC - AC$.
 (b) If a triangle is inscribed in a circle, then the product of any two sides of the triangle is equal to the product of the diameter and the perpendicular distance of the third side from the opposite vertex. Prove the above statement. (1979)
3. (a) A balloon is observed simultaneously from three points A , B and C on a straight road directly beneath it. The angular elevation at B is twice that at A and the angular elevation at C is thrice that at A . If the distance between A and B is a and the distance between B and C is b , find the height of the balloon in terms of a and b .
 (b) Find the area of the smaller part of a disc of radius 10 cm, cut off by a chord AB which subtends an angle of $22\frac{1}{2}^\circ$ at the circumference. (1979)
4. ABC is a triangle. D is the middle point of BC . If AD is perpendicular to AC , then prove that (1980)
- $$\cos A \cos C = \frac{2(c^2 - a^2)}{3ac}$$
5. ABC is a triangle with $AB = AC$. D is any point on the side BC . E and F are points on the side AB and AC , respectively, such that DE is parallel to AC , and DF is parallel to AB . Prove that (1980)
- $$DF + FA + AE + ED = AB + AC$$
6. (i) PQ is a vertical tower. P is the foot and Q is the top of the tower. A , B , C are three points in the horizontal plane through P . The angles of elevation of Q from A , B , C are equal, and each is equal to θ . The sides of the triangle ABC are a , b , c ; and the area of the triangle ABC is Δ . Show that the height of the tower is $\frac{abc \tan \theta}{4\Delta}$.
 (ii) AB is a vertical pole. The end A is on the level ground. C is the middle point of AB . P is a point on the level ground. The portion CB subtends an angle β at P . If $AP = n AB$, then show that $\tan \beta = \frac{n}{2n^2 + 1}$ (1980)
7. Let the angles A , B , C of a triangle ABC be in A.P. and let $b : c = \sqrt{3} : \sqrt{2}$. Find the angle A . (1981 - 2 Marks)

8. A vertical pole stands at a point Q on a horizontal ground. A and B are points on the ground, d meters apart. The pole subtends angles α and β at A and B respectively. AB subtends an angle γ at Q . Find the height of the pole. (1982 - 3 Marks)
9. Four ships A, B, C and D are at sea in the following relative positions : B is on the straight line segment AC , B is due North of D and D is due west of C . The distance between B and D is 2 km. $\angle BDA = 40^\circ$, $\angle BCD = 25^\circ$. What is the distance between A and D ? [Take $\sin 25^\circ = 0.423$] (1983 - 3 Marks)
10. The ex-radii r_1, r_2, r_3 of $\triangle ABC$ are in H.P. Show that its sides a, b, c are in A.P. (1983 - 3 Marks)
11. For a triangle ABC it is given that $\cos A + \cos B + \cos C = \frac{3}{2}$. Prove that the triangle is equilateral. (1984 - 4 Marks)
12. With usual notation, if in a triangle ABC ;
 $\frac{b+c}{11} = \frac{c+a}{12} = \frac{a+b}{13}$ then prove that $\frac{\cos A}{7} = \frac{\cos B}{19} = \frac{\cos C}{25}$. (1984 - 4 Marks)
13. A ladder rests against a wall at an angle α to the horizontal. Its foot is pulled away from the wall through a distance a , so that it slides a distance b down the wall making an angle β with the horizontal. Show that $a = b \tan \frac{1}{2}(\alpha + \beta)$ (1985 - 5 Marks)
14. In a triangle ABC , the median to the side BC is of length $\frac{1}{\sqrt{11-6\sqrt{3}}}$ (1985 - 5 Marks)
 and it divides the angle A into angles 30° and 45° . Find the length of the side BC .
15. If in a triangle ABC , $\cos A \cos B + \sin A \sin B \sin C = 1$, Show that $a : b : c = 1 : 1 : \sqrt{2}$ (1986 - 5 Marks)
16. A sign-post in the form of an isosceles triangle ABC is mounted on a pole of height h fixed to the ground. The base BC of the triangle is parallel to the ground. A man standing on the ground at a distance d from the sign-post finds that the top vertex A of the triangle subtends an angle β and either of the other two vertices subtends the same angle α at his feet. Find the area of the triangle. (1988 - 5 Marks)
17. ABC is a triangular park with $AB = AC = 100$ m. A television tower stands at the midpoint of BC . The angles of elevation of the top of the tower at A, B, C are $45^\circ, 60^\circ, 60^\circ$, respectively. Find the height of the tower. (1989 - 5 Marks)
18. A vertical tower PQ stands at a point P . Points A and B are located to the South and East of P respectively. M is the mid point of AB . PAM is an equilateral triangle; and N is the foot of the perpendicular from P on AB . Let $AN = 20$ metres and the angle of elevation of the top of the tower at N is $\tan^{-1}(2)$. Determine the height of the tower and the angles of elevation of the top of the tower at A and B . (1990 - 4 Marks)
19. The sides of a triangle are three consecutive natural numbers and its largest angle is twice the smallest one. Determine the sides of the triangle. (1991 - 4 Marks)
20. In a triangle of base a the ratio of the other two sides is $r (< 1)$. Show that the altitude of the triangle is less than of equal to $\frac{ar}{1-r^2}$ (1991 - 4 Marks)
21. A man notices two objects in a straight line due west. After walking a distance c due north he observes that the objects subtend an angle α at his eye; and, after walking a further distance $2c$ due north, an angle β . Show that the distance between the objects is $\frac{8c}{3 \cot \beta - \cot \alpha}$; the height of the man is being ignored. (1991 - 4 Marks)
22. Three circles touch the one another externally. The tangent at their point of contact meet at a point whose distance from a point of contact is 4. Find the ratio of the product of the radii to the sum of the radii of the circles. (1992 - 4 Marks)
23. An observer at O notices that the angle of elevation of the top of a tower is 30° . The line joining O to the base of the tower makes an angle of $\tan^{-1}(1/\sqrt{2})$ with the North and is inclined Eastwards. The observer travels a distance of 300 meters towards the North to a point A and finds the tower to his East. The angle of elevation of the top of the tower at A is ϕ , Find ϕ and the height of the tower (1993 - 5 Marks)
24. A tower AB leans towards west making an angle α with the vertical. The angular elevation of B , the topmost point of the tower is β as observed from a point C due west of A at a distance d from A . If the angular elevation of B from a point D due east of C at a distance $2d$ from C is γ , then prove that $2 \tan \alpha = -\cot \beta + \cot \gamma$. (1994 - 4 Marks)
25. Let $A_1, A_2, \dots, A_n$ be the vertices of an n -sided regular polygon such that $\frac{1}{A_1 A_2} = \frac{1}{A_1 A_3} + \frac{1}{A_1 A_4}$, Find the value of n . (1994 - 4 Marks)
26. Consider the following statements concerning a triangle ABC (1994 - 5 Marks)
- (i) The sides a, b, c and area Δ are rational.
- (ii) $a, \tan \frac{B}{2}, \tan \frac{C}{2}$ are rational.
- (iii) $a, \sin A, \sin B, \sin C$ are rational.
- Prove that (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i)
27. A bird flies in a circle on a horizontal plane. An observer stands at a point on the ground. Suppose 60° and 30° are the maximum and the minimum angles of elevation of the bird and that they occur when the bird is at the points P and Q respectively on its path. Let θ be the angle of elevation of the bird when it is a point on the arc of the circle exactly midway between P and Q . Find the numerical value of $\tan^2 \theta$. (Assume that the observer is not inside the vertical projection of the path of the bird.) (1998 - 8 Marks)

Properties of Triangle

28. Prove that a triangle ABC is equilateral if and only if
 $\tan A + \tan B + \tan C = 3\sqrt{3}$. (1998 - 8 Marks)
29. Let ABC be a triangle having O and I as its circumcenter and in centre respectively. If R and r are the circumradius and the inradius, respectively, then prove that $(IO)^2 = R^2 - 2Rr$. Further show that the triangle BIO is a right-angled triangle if and only if b is arithmetic mean of a and c . (1999 - 10 Marks)
30. Let ABC be a triangle with incentre I and inradius r . Let D, E, F be the feet of the perpendiculars from I to the sides BC, CA and AB respectively. If r_1, r_2 and r_3 are the radii of circles inscribed in the quadrilaterals $AFIE, BDIF$ and $CEID$ respectively, prove that
- $$\frac{r_1}{r-r_1} + \frac{r_2}{r-r_2} + \frac{r_3}{r-r_3} = \frac{r_1 r_2 r_3}{(r-r_1)(r-r_2)(r-r_3)}.$$
- (2000 - 7 Marks)
31. If Δ is the area of a triangle with side lengths a, b, c , then show that $\Delta \leq \frac{1}{4}\sqrt{(a+b+c)abc}$. Also show that the equality occurs in the above inequality if and only if $a = b = c$. (2001 - 6 Marks)

32. If I_n is the area of n sided regular polygon inscribed in a circle of unit radius and O_n be the area of the polygon circumscribing the given circle, prove that

$$I_n = \frac{O_n}{2} \left(1 + \sqrt{1 - \left(\frac{2I_n}{n} \right)^2} \right). \quad (2003 - 4 Marks)$$

I Integer Value Correct Type

1. Let ABC and ABC' be two non-congruent triangles with sides $AB = 4, AC = AC' = 2\sqrt{2}$ and angle $B = 30^\circ$. The absolute value of the difference between the areas of these triangles is (2009)
2. Consider a triangle ABC and let a, b and c denote the lengths of the sides opposite to vertices A, B and C respectively. Suppose $a=6, b=10$ and the area of the triangle is $15\sqrt{3}$, if $\angle ACB$ is obtuse and if r denotes the radius of the incircle of the triangle, then r^2 is equal to (2010)

Section-B

JEE Main / AIEEE

1. The sides of a triangle are $3x+4y, 4x+3y$ and $5x+5y$ where $x, y > 0$ then the triangle is [2002]
 (a) right angled (b) obtuse angled
 (c) equilateral (d) none of these
2. In a triangle with sides $a, b, c, r_1 > r_2 > r_3$ (which are the ex-radii) then [2002]
 (a) $a > b > c$ (b) $a < b < c$
 (c) $a > b$ and $b < c$ (d) $a < b$ and $b > c$
3. The sum of the radii of inscribed and circumscribed circles for an n sided regular polygon of side a , is [2003]
 (a) $\frac{a}{4} \cot\left(\frac{\pi}{2n}\right)$ (b) $a \cot\left(\frac{\pi}{n}\right)$
 (c) $\frac{a}{2} \cot\left(\frac{\pi}{2n}\right)$ (d) $a \cot\left(\frac{\pi}{2n}\right)$
4. In a triangle ABC , medians AD and BE are drawn. If $AD = 4, \angle DAB = \frac{\pi}{6}$ and $\angle ABE = \frac{\pi}{3}$, then the area of the ΔABC is [2003]
 (a) $\frac{64}{3}$ (b) $\frac{8}{3}$ (c) $\frac{16}{3}$ (d) $\frac{32}{3\sqrt{3}}$
5. If in a $\Delta ABC, a \cos^2\left(\frac{C}{2}\right) + c \cos^2\left(\frac{A}{2}\right) = \frac{3b}{2}$, then the sides a, b and c [2003]
 (a) satisfy $a + b = c$ (b) are in A.P
 (c) are in G.P (d) are in H.P.
6. The sides of a triangle are $\sin \alpha, \cos \alpha$ and $\sqrt{1 + \sin \alpha \cos \alpha}$ for some $0 < \alpha < \frac{\pi}{2}$. Then the greatest angle of the triangle is [2004]
 (a) 150° (b) 90° (c) 120° (d) 60°
7. A person standing on the bank of a river observes that the angle of elevation of the top of a tree on the opposite bank of the river is 60° and when he retires 40 meters away from the tree the angle of elevation becomes 30° . The breadth of the river is [2004]
 (a) 60 m (b) 30 m (c) 40 m (d) 20 m
8. In a triangle ABC , let $\angle C = \frac{\pi}{2}$. If r is the inradius and R is the circumradius of the triangle ABC , then $2(r + R)$ equals [2005]
 (a) $b + c$ (b) $a + b$ (c) $a + b + c$ (d) $c + a$
9. If in a ΔABC , the altitudes from the vertices A, B, C on opposite sides are in H.P, then $\sin A, \sin B, \sin C$ are in [2005]
 (a) G.P. (b) A.P. (c) A.P-G.P. (d) H.P
10. A tower stands at the centre of a circular park. A and B are two points on the boundary of the park such that $AB (= a)$ subtends an angle of 60° at the foot of the tower, and the angle of elevation of the top of the tower from A or B is 30° . The height of the tower is [2007]
 (a) $a/\sqrt{3}$ (b) $a\sqrt{3}$ (c) $2a/\sqrt{3}$ (d) $2a\sqrt{3}$.

11. AB is a vertical pole with B at the ground level and A at the top. A man finds that the angle of elevation of the point A from a certain point C on the ground is 60° . He moves away from the pole along the line BC to a point D such that $CD=7$ m. From D the angle of elevation of the point A is 45° . Then the height of the pole is [2008]
- (a) $\frac{7\sqrt{3}}{2} \frac{1}{\sqrt{3}-1} m$ (b) $\frac{7\sqrt{3}}{2} (\sqrt{3}+1)m$
- (c) $\frac{7\sqrt{3}}{2} (\sqrt{3}-1)m$ (d) $\frac{7\sqrt{3}}{2} \frac{1}{\sqrt{3}+1} m$
12. For a regular polygon, let r and R be the radii of the inscribed and the circumscribed circles. A **false** statement among the following is [2010]
- (a) There is a regular polygon with $\frac{r}{R} = \frac{1}{\sqrt{2}}$
- (b) There is a regular polygon with $\frac{r}{R} = \frac{2}{3}$
- (c) There is a regular polygon with $\frac{r}{R} = \frac{\sqrt{3}}{2}$
- (d) There is a regular polygon with $\frac{r}{R} = \frac{1}{2}$
13. A bird is sitting on the top of a vertical pole 20 m high and its elevation from a point O on the ground is 45° . It flies off horizontally straight away from the point O . After one second, the elevation of the bird from O is reduced to 30° . Then the speed (in m/s) of the bird is [JEE M 2014]
- (a) $20\sqrt{2}$ (b) $20(\sqrt{3}-1)$
- (c) $40(\sqrt{2}-1)$ (d) $40(\sqrt{3}-\sqrt{2})$
14. If the angles of elevation of the top of a tower from three collinear points A , B and C , on a line leading to the foot of the tower, are 30° , 45° and 60° respectively, then the ratio, $AB : BC$, is : [JEE M 2015]
- (a) $1 : \sqrt{3}$ (b) $2 : 3$
- (c) $\sqrt{3} : 1$ (d) $\sqrt{3} : \sqrt{2}$

13

Properties of Triangle

Section-A : JEE Advanced/ IIT-JEE

- A** 1. BC 2. AP^2 3. $\angle B$ 4. $(5, \infty)$ 5. arithmetic 6. $\operatorname{cosec} \frac{\pi}{9}$
7. $\frac{\sqrt{3}+1}{2}$ sq. units 8. 90° 9. 113° 10. $\frac{a^2}{6}$ sq. units 11. $16:7$
- C** 1. (c) 2. (b) 3. (c) 4. (c) 5. (a) 6. (b)
 7. (a) 8. (b) 9. (b) 10. (d) 11. (a) 12. (d)
 13. (b) 14. (b) 15. (c) 16. (b) 17. (d) 18. (c)
 19. (b)
- D** 1. (a, d) 2. (a, d) 3. (b) 4. (c) 5. (a, b, c, d) 6. (b)
 7. (b, d) 8. (a, c, d)
- E** 1. $8\sqrt{\frac{17}{5}}$ 3. (a) $\frac{a}{2b}\sqrt{(a+b)(3b-a)}$ (b) 3.91 sq. cm 7. 75°
8. $\frac{d}{\sqrt{\cot^2 \alpha + \cot^2 \beta - \cot \alpha \cot \beta \cot \gamma}}$ 9. 4.28 km 14. 2 units
16. $(d \tan \beta - h)\sqrt{h^2 \cot^2 \alpha - d^2}$ 17. $50\sqrt{3} \text{ m}$ 18. $40\sqrt{3} \text{ m}, 60^\circ, 45^\circ$ 19. 4, 5, 6
22. 16:1 23. $45^\circ, 150\sqrt{2} \text{ m}$ 25. 7 27. $\frac{3}{5}$
- I** 1. 4 2. 3

Section-B : JEE Main/ AIEEE

1. (b) 2. (a) 3. (c) 4. (d) 5. (b) 6. (c)
 7. (d) 8. (b) 9. (b) 10. (a) 11. (b) 12. (b)
 13. (b) 14. (c)

Section-A

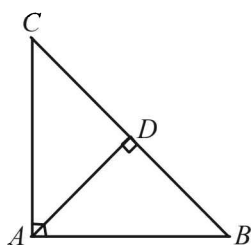
JEE Advanced/ IIT-JEE

A. Fill in the Blanks

1. We know that altitude from right vertex to hypotenuse in right angled triangle divides it into two triangles each being similar to the original triangle.

$$\therefore \triangle BDA \sim \triangle BAC$$

$$\Rightarrow \frac{BD}{BA} = \frac{AB}{BC}$$



2. In $\triangle APQ$ and $\triangle ACB$

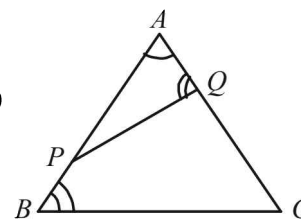
$$\angle A = \angle A \text{ (common)}$$

$$\angle AQP = \angle ABC \text{ (given)}$$

$$\therefore \triangle APQ \sim \triangle ACB \text{ fig}$$

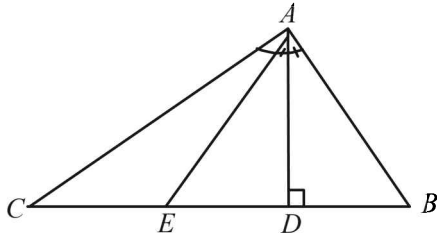
is with (iii) (by AA similarity)

$$\Rightarrow \frac{Ar(\triangle APQ)}{Ar(\triangle ACB)} = \frac{AP^2}{AC^2}$$



Properties of Triangle

3. We have $\angle BAE = \angle CAE$ (given)
and $\angle ADB = \angle ADC = 90^\circ$ (given)



Now $\angle DAE = \angle BAE - \angle BAD$
 $= \angle CAE - (90^\circ - \angle B)$
 $= (\angle CAD - \angle DAE) - 90^\circ + \angle B$
 $= (90^\circ - \angle C) - \angle DAE - 90^\circ + \angle B$

$$\Rightarrow 2\angle DAE = \angle B - \angle C \Rightarrow \angle DAE = \frac{1}{2}(\angle B - \angle C)$$

4. If $a^2 + 2a$, $2a + 3$, $a^2 + 3a + 8$ are sides of a Δ then sum of any two sides is greater than the third side.

Let $x = a^2 + 2a$; $y = 2a + 3$; $z = a^2 + 3a + 8$

Then $x + y > z \Rightarrow a^2 + 4a + 3 > a^2 + 3a + 8$

$$\Rightarrow a > 5 \quad \dots(1)$$

$$y + z > x$$

$$\Rightarrow a^2 + 5a + 11 > a^2 + 2a \Rightarrow 3a > -11$$

$$\Rightarrow a > -11/3 \quad \dots(2)$$

$$z + x > y$$

$$\Rightarrow 2a^2 + 5a + 8 > 2a + 3 \Rightarrow 2a^2 + 3a + 5 > 0$$

Here coeff. of $a^2 > 0$ and $D = 9 - 40 = -ve$

$\therefore$ it is true for all values of a . Therefore, identity.

Combining (1) and (2), we get $a > 5$.

$$\therefore a \in (5, \infty)$$

5. $\cot A$, $\cot B$, $\cot C$ are in A.P.

$$\Rightarrow \cot B - \cot A = \cot C - \cot B$$

$$\Rightarrow \frac{\sin(A-B)}{\sin A \sin B} = \frac{\sin(B-C)}{\sin B \sin C}$$

$$\Rightarrow \sin(A-B) \sin(A+B) = \sin(B+C) \sin(B-C)$$

$$\Rightarrow \sin^2 A - \sin^2 B = \sin^2 B - \sin^2 C$$

$$\Rightarrow a^2 - b^2 = b^2 - c^2 \Rightarrow a^2, b^2, c^2 \text{ are in A.P.}$$

6. Let $AB = 2$ units be one of the sides of the polygon.

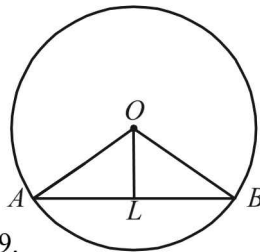
Then $\angle AOB = 2\pi/9$ where O is the centre of circle.

If $OL \perp AB$, then $AL = 1$

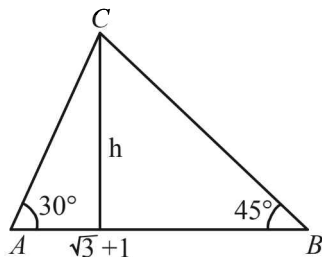
and $\angle AOL = \pi/9$

$\therefore$ Radius of the circle

$$= OA = AL \operatorname{cosec} \pi/9 = \operatorname{cosec} \pi/9.$$



- 7.



$$\tan 30^\circ = \frac{h}{\sqrt{3} + 1 - h} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow h\sqrt{3} + h = \sqrt{3} + 1 \Rightarrow h = 1$$

$$\therefore \text{Area} = \frac{1}{2} \times (\sqrt{3} + 1) \times 1 = \frac{\sqrt{3} + 1}{2} \text{ sq. units}$$

8. In ΔABC , we have

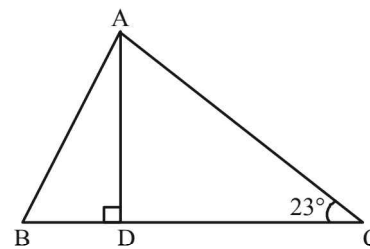
$$\frac{2 \cos A}{a} + \frac{\cos B}{b} + \frac{2 \cos C}{c} = \frac{a}{bc} + \frac{b}{ca}$$

$$\Rightarrow \frac{2 \left(\frac{b^2 + c^2 - a^2}{2bc} \right)}{a} + \frac{\frac{a^2 + c^2 - b^2}{2ac}}{b} + \frac{2 \left(\frac{b^2 + a^2 - c^2}{2ab} \right)}{c} = \frac{a^2 + b^2}{abc}$$

$$\Rightarrow 2b^2 + 2c^2 - 2a^2 + a^2 + c^2 - b^2 + 2a^2 + 2b^2 - 2c^2 = 2a^2 + 2b^2$$

$$\Rightarrow b^2 + c^2 = a^2 \Rightarrow \Delta \text{ is right } \angle \text{ed' at } A \Rightarrow \angle A = 90^\circ$$

- 9.



$$AD = b \sin 23^\circ \text{ but given } AD = \frac{abc}{b^2 - c^2}$$

$$\Rightarrow b \sin 23^\circ = \frac{abc}{b^2 - c^2} \Rightarrow \frac{\sin 23^\circ}{c} = \frac{a}{b^2 - c^2} \quad \dots(1)$$

$$\text{Also } \frac{\sin 23^\circ}{c} = \sin A \quad \dots(2)$$

(by sine law)

From (1) and (2)

$$\frac{a}{b^2 - c^2} = \frac{\sin A}{a} \Rightarrow \frac{a^2}{b^2 - c^2} = \frac{\sin^2 A}{\sin^2 B - \sin^2 C} = \sin A$$

$$\Rightarrow \sin(B+C) \sin(B-C) = \sin A$$

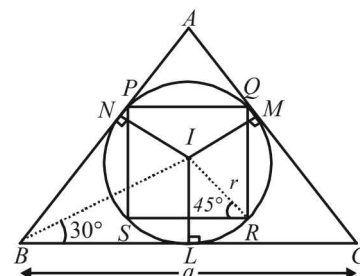
$$\Rightarrow \sin(B-C) = 1 \Rightarrow B-C = 90^\circ \Rightarrow B = 90^\circ + 23^\circ = 113^\circ$$

10. Given that ABC is an equilateral Δ , of side a , r is the radius of circle inscribed in it.

$$\therefore \text{ In } \Delta IBL, \tan 30^\circ = \frac{r}{a/2} \Rightarrow r = \frac{a}{2\sqrt{3}}$$

If $PQRS$ is the square inscribed in circle of radius r , then

$$\text{side of square} = 2(r \cos 45^\circ) = \frac{2r}{\sqrt{2}} = \frac{2}{\sqrt{2}} \times \frac{a}{2\sqrt{3}} = \frac{a}{\sqrt{6}}$$



$$\therefore \text{ area of square} = a^2/6 \text{ sq. units.}$$

11. Let
- $a = 4k$
- ,
- $b = 5k$
- ,
- $c = 6k$

$$s = \frac{15}{2}k, s - a = \frac{7}{2}k, s - b = \frac{5}{2}k, s - c = \frac{3}{2}k$$

$$S^2 = 15 \times 7 \times 5 \times 3 \left(\frac{k}{2}\right)^4 \Rightarrow S = 15\sqrt{7} \left(\frac{k}{2}\right)^2$$

$$r = \frac{S}{s} = 15\sqrt{7} \left(\frac{k}{2}\right)^4 \div \frac{15}{2}k = \sqrt{7} \frac{k}{2} \quad R = \frac{abc}{4S}$$

$$= \frac{4.5 \cdot 6k^3}{4 \cdot 15\sqrt{7}k^2/4} = \frac{8}{\sqrt{7}}k.$$

$$\therefore \frac{R}{r} = \frac{8}{\sqrt{7}} \div \frac{\sqrt{7}}{2} = \frac{16}{7}$$

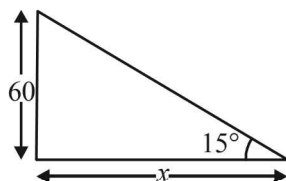
C. MCQs with ONE Correct Answer

1. (c)
- $QS : SR = PQ : PR$
- (as bisector of an angle, in a triangle, divides the opposite side in the same ratio as the sides containing the angle.)

2. (b) $\tan 15^\circ = \frac{60}{x}$

$$\Rightarrow x = 60 \cot 15^\circ$$

$$= 60 \left[\frac{\sqrt{3}+1}{\sqrt{3}-1} \right]$$



3. (c) Given that
- $A > B$
-
- and
- $3 \sin x - 4 \sin^3 x - k = 0$
- ,
- $0 < k < 1$

$$\Rightarrow \sin 3x = k$$

As A and B satisfy above eq. (given)

$$\therefore \sin 3A = k, \sin 3B = k$$

$$\Rightarrow \sin 3A - \sin 3B = 0$$

$$\Rightarrow 2 \cos \frac{3A+3B}{2} \sin \frac{3A-3B}{2} = 0$$

$$\Rightarrow \cos \left(\frac{3A+3B}{2} \right) = 0 \text{ or } \sin \left(\frac{3A-3B}{2} \right) = 0$$

$$\Rightarrow \frac{3A+3B}{2} = 90^\circ \text{ or } \frac{3A-3B}{2} = 0$$

$$\Rightarrow A+B=60^\circ \text{ or } A=B$$

But given that $A > B$, $\therefore A \neq B$

$$\text{Thus, } A+B=60^\circ$$

$$\text{But } A+B+C=180^\circ \Rightarrow C=180^\circ-60^\circ=120^\circ$$

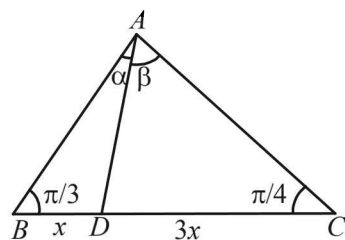
$$\therefore C=2\pi/3$$

4. (c) Let
- $a=3, b=5, c=7$
- then the largest
- $\angle$
- is opp. to the longest side, i.e.,
- $\angle C$

$$\therefore \cos C = \frac{a^2+b^2-c^2}{2ab} = \frac{9+25-49}{2 \times 3 \times 5} = \frac{-1}{2}$$

$$\Rightarrow C=2\pi/3$$

5. (a)

In $\triangle ABD$, applying Sine law we get

$$\frac{AD}{\sin \pi/3} = \frac{x}{\sin \alpha}$$

$$\Rightarrow AD = \frac{\sqrt{3}x}{2 \sin \alpha} \quad \dots(1)$$

In $\triangle ACD$, applying Sine law, we get

$$\frac{AD}{\sin \pi/4} = \frac{3x}{\sin \beta}$$

$$\Rightarrow AD = \frac{3x}{\sqrt{2} \sin \beta} \quad \dots(2)$$

$$\text{From (i) and (ii) } \frac{\sqrt{3}x}{2 \sin \alpha} = \frac{3x}{\sqrt{2} \sin \beta}$$

$$\Rightarrow \frac{\sin \alpha}{\sin \beta} = \frac{1}{\sqrt{6}}$$

6. (b) We know that
- $A+B+C=180^\circ$

$$\Rightarrow A+C-B=180^\circ-2B$$

$$\text{Now } 2ac \sin \left[\frac{1}{2}(A-B+C) \right] = 2ac \sin(90^\circ-B)$$

$$= 2ac \cos B = \frac{2ac(a^2+c^2-b^2)}{2ac} = a^2+c^2-b^2$$

7. (a) We know by Sine rule

$$\frac{c}{\sin C} = 2R \Rightarrow C = 2R \sin C$$

$$\Rightarrow C = 2R \quad (\because \angle C = 90^\circ)$$

$$\text{Also } \tan \frac{C}{2} = \frac{r}{s-c}$$

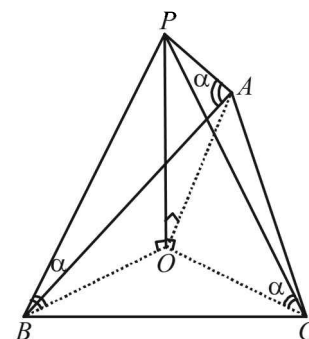
$$\Rightarrow r = s - c \quad (\because \angle C = 90^\circ)$$

$$\Rightarrow a+b-c=2r$$

$$\text{or } 2r+c=a+b$$

$$\text{or } 2r+2R=a+b \quad (\text{Using } C=2R)$$

8. (b)

Let $OP = \text{Pole}$,

$$\angle PAO = \angle PBO = \angle PCQ = \alpha$$

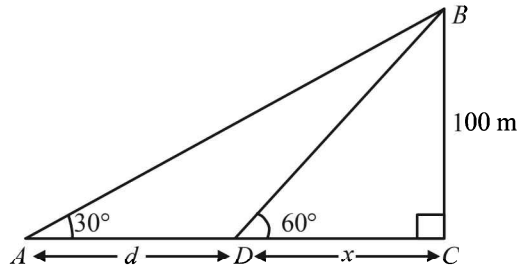
$$\frac{OP}{OB} = \tan \alpha \Rightarrow OB = OP \cot \alpha \quad \dots(1)$$

$$\text{Similarly } OA = OP \cot \alpha \quad \dots(2)$$

$$\text{Similarly } OC = OP \cot \alpha \quad \dots(3)$$

From (1), (2) and (3), $OA = OB = OC \Rightarrow O$ is the point of circumcentre of the triangle ABC .

9. (b)



$$\text{In } \triangle CBD, \tan 60^\circ = \frac{100}{x} \Rightarrow x = \frac{100}{\sqrt{3}}$$

$$\text{In } \triangle ACB, \tan 30^\circ = \frac{100}{x+d} \Rightarrow x+d = 100\sqrt{3}$$

$$\Rightarrow d = 100\sqrt{3} - \frac{100}{\sqrt{3}} = \frac{200}{\sqrt{3}} = \frac{200\sqrt{3}}{3} \text{ m}$$

10. (d) We know by Sine law in $\triangle ABC$ as

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin(\pi - A - B)} = 2R$$

$$\Rightarrow \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin(A+B)} = 2R$$

(a) If we know a , $\sin A$, $\sin B$ we can find b , c ; values of $\angle$'s A , B and C all.

(b) Using a , b , c we can find $\angle A$, $\angle B$, $\angle C$ using cosine law.

(c) a , $\sin B$, R are given then $\sin A$, b and hence $\sin(A+B)$ and then C can be found.

(d) If we know a , $\sin A$, R then we know only the ratio $\frac{b}{\sin B} = \frac{c}{\sin(A+B)}$; we can not determine

the values of b , c , $\sin B$, $\sin C$ separately.
 $\therefore \Delta$ can not be determined in this case.

11. (a) Given that $A : B : C = 4 : 1 : 1$

$$\text{Let } A = 4x, B = x, C = x$$

$$\text{But } A + B + C = 180^\circ$$

$$\Rightarrow 4x + x + x = 180^\circ \Rightarrow x = 30^\circ$$

$$\therefore A = 120^\circ, B = 30^\circ, C = 30^\circ$$

$$\text{By sine law, } \frac{a}{\sin 120^\circ} = \frac{b}{\sin 30^\circ} = \frac{c}{\sin 30^\circ}$$

$$\Rightarrow \frac{a}{\sqrt{3}/2} = \frac{b}{1/2} = \frac{c}{1/2} \Rightarrow a : b : c = \sqrt{3} : 1 : 1$$

$$\therefore \text{Ratio of longest side to the perimeter} \\ = \sqrt{3} : 1 + 1 + \sqrt{3} = \sqrt{3} : 2 + \sqrt{3}$$

12. (d) Sides are in the Ratio $1 : \sqrt{3} : 2$

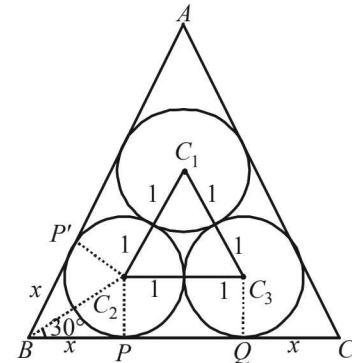
$$\text{Let } a = k, b = \sqrt{3}k \text{ and } c = 2k$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{\sqrt{3}}{2} \Rightarrow A = \frac{\pi}{6}$$

$$\cos B = \frac{c^2 + a^2 - b^2}{2ac} = \frac{1}{2} \Rightarrow B = \frac{\pi}{3}$$

$$\Rightarrow C = \pi - (A + B) = \frac{\pi}{2} \Rightarrow A : B : C = 1 : 2 : 3$$

13. (b) The situation is as shown in the figure. For circle with centre C_2 , BP and BP' are two tangents from B to circle, therefore BC_2 must be the $\angle$ bisector of $\angle B$. But $\angle B = 60^\circ (\because \triangle ABC \text{ is an equilateral } \Delta)$



$$\therefore \angle C_2BP = 30^\circ$$

$$\therefore \tan 30^\circ = \frac{1}{x} \Rightarrow x = \sqrt{3}$$

$$\therefore BC = BP + PQ + QC = x + 2 + x = 2 + 2\sqrt{3}$$

$$\therefore \text{Area of } \triangle ABC = \frac{\sqrt{3}}{4} \times (2 + 2\sqrt{3})^2 \\ = 4\sqrt{3} + 6 \text{ sq. units.}$$

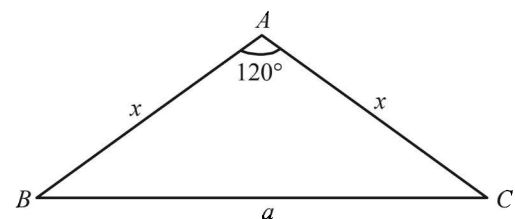
14. (b) Let us consider $\frac{b-c}{a}$, which is involved in each of the these options.

$$\frac{b-c}{a} = \frac{\sin B - \sin C}{\sin A} = \frac{2 \cos \left(\frac{B+C}{2} \right) \sin \left(\frac{B-C}{2} \right)}{2 \sin A / 2 \cos A / 2} \\ = \frac{\sin A / 2 \sin \left(\frac{B-C}{2} \right)}{\sin A / 2 \cos A / 2} = \frac{\sin \left(\frac{B-C}{2} \right)}{\cos A / 2}$$

$$\therefore (b-c) \cos A / 2 = a \sin \left(\frac{B-C}{2} \right)$$

15. (c) By Sine law in $\triangle ABC$

$$\frac{x}{\sin 30^\circ} = \frac{a}{\sin 120^\circ} \Rightarrow a = x\sqrt{3}$$



$$\therefore \Delta = \frac{1}{2} \times x \times x \sin 120^\circ = \frac{\sqrt{3}}{4} x^2$$

$$\text{Also, } \sqrt{3} = \frac{\Delta}{s} \Rightarrow \frac{(2x+a)}{2} \sqrt{3} = \frac{\sqrt{3}}{4} x^2$$

$$\Rightarrow x = 2(2 + \sqrt{3}) \Rightarrow \Delta = \frac{\sqrt{3}}{4} \times 4(4 + 3 + 4\sqrt{3})$$

$$\Rightarrow \Delta = 7\sqrt{3} + 12 \text{ sq. units.}$$

16. (b) Given $AB \parallel CD$, $CD = 2AB$ Let $AB = a$ then $CD = 2a$
Let radius of circle be r . Let circle touches AB at P , BC at Q , AD at R and CD at S .

Then $AR = AP = r$, $BP = BQ = a - r$

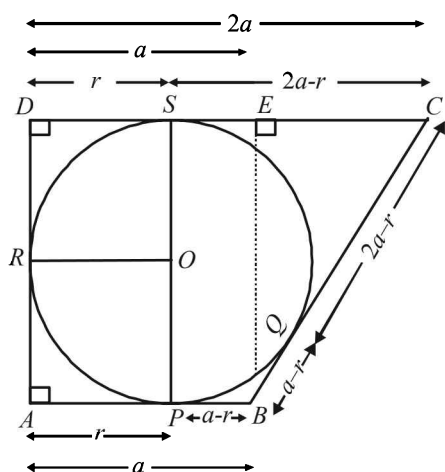
$DR = DS = r$ and $CQ = CS = 2a - r$ In $\triangle BEC$

$$BC^2 = BE^2 + EC^2 \Rightarrow (a - r + 2a - r)^2 = (2r)^2 + (a)^2$$

$$\Rightarrow 9a^2 + 4r^2 - 12ar = 4r^2 + a^2$$

$$\Rightarrow a = \frac{3}{2}r \quad \dots(1)$$

Also Ar (quad. $ABCD$) = 18



$$\Rightarrow a \times 2r + \frac{1}{2} \times a \times 2r = 18$$

$$\Rightarrow ar = 6 \Rightarrow \frac{3r^2}{2} = 6 \text{ (using equation (1))}$$

$$\Rightarrow r^2 = 4 \Rightarrow r = 2$$

17. (d) $\therefore A, B, C$ are in A.P.

$$\therefore B = 60^\circ \text{ then } \frac{a}{c} \sin 2C + \frac{c}{a} \sin 2A$$

$$= \frac{K \sin A}{K \sin C} \cdot 2 \sin C \cos C + \frac{K \sin C}{K \sin A} \cdot \sin A \cos A$$

$$= 2(\sin A \cos C + \cos A \sin C) = 2 \sin(A + C) = 2 \sin B$$

$$= 2 \times \frac{\sqrt{3}}{2} = \sqrt{3}$$

18. (c) We have,

$$\frac{2 \sin P - \sin 2P}{2 \sin P + \sin 2P} = \frac{2 \sin P - 2 \sin P \cos P}{2 \sin P + 2 \sin P \cos P} = \frac{1 - \cos P}{1 + \cos P}$$

$$= \frac{2 \sin^2 \frac{P}{2}}{2 \cos^2 \frac{P}{2}} = \tan^2 \frac{P}{2} = \frac{(s-b)(s-c)}{s(s-a)}$$

$$\text{where } s = \frac{a+b+c}{2}$$

$$= \frac{(s-b)^2(s-c)^2}{s(s-a)(s-b)(s-c)} = \frac{(a+c-b)^2(a+b-c)^2}{16\Delta^2}$$

$$= \frac{1 \times 9}{16\Delta^2} = \left(\frac{3}{4\Delta}\right)^2$$

19. (b) Let two sides of Δ be a and b .

Then $a + b = x$ and $ab = y$

Also given $x^2 - c^2 = y$, where c is the third side of Δ .

$$\Rightarrow (a+b)^2 - c^2 = ab \Rightarrow a^2 + b^2 - c^2 = -ab$$

$$\Rightarrow \frac{a^2 + b^2 - c^2}{2ab} = -\frac{1}{2} \Rightarrow \cos c = -\frac{1}{2} \Rightarrow c = 120^\circ$$

$$\therefore \frac{r}{R} = \frac{\Delta}{s} \times \frac{4\Delta}{abc} \text{ where } \Delta = \text{area of triangle}$$

$$\Rightarrow \frac{r}{R} = \frac{4\Delta^2}{(a+b+c)abc} = \frac{8 \times \left(\frac{1}{2} ab \sin c\right)^2}{(a+b+c)abc}$$

$$\Rightarrow \frac{2a^2b^2 \sin^2 120^\circ}{(a+b+c)abc} = \frac{2ab \times \frac{3}{4}}{(x+c)c} = \frac{3y}{2c(x+c)}$$

D. MCQs with ONE or More THAN One Correct

1. (a, d) We have

$$\frac{\sin A}{a} = \frac{\sin B}{b} \Rightarrow a \sin B = b \sin A$$

$$(a) \quad b \sin A = a \Rightarrow a \sin B = a$$

$$\Rightarrow \sin B = 1 \Rightarrow B = \pi/2$$

Since $A < \pi/2$, the $\triangle ABC$ is possible.

$$(b) \quad b \sin A > a \Rightarrow a \sin B > a \Rightarrow \sin B > 1$$

Which is impossible. Hence the possibility (b) is ruled out.

Similarly (c) can be shown to be impossible.

$$(d) \quad b \sin A < a \Rightarrow a \sin B < a \Rightarrow \sin B < 1 \text{ so value of } \angle B \text{ exists.}$$

Now, $b > a \Rightarrow B > A$. Since $A < \pi/2$

The $\triangle ABC$ is possible when $B > \pi/2$ or $B < \pi/2$.

$$(e) \quad \text{Since } b = a, \text{ we have } B = A. \text{ But } A > \pi/2$$

$$\therefore B > \pi/2. \text{ But this is not possible for any triangle.}$$

2. (a, d) Since the angles of triangle are in A.P., Let

$$\angle A = x - d, \angle B = x, \angle C = x + d$$

Then by $\angle$ sum property of triangle, we have

$$\angle A + \angle B + \angle C = 180^\circ$$

$$\Rightarrow x - d + x + x + d = 180^\circ$$

$$\Rightarrow 3x = 180^\circ \Rightarrow x = 60^\circ$$

$$\therefore \angle B = 60^\circ$$

$$\text{Using cosine formula } \cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\text{We get } \cos 60^\circ = \frac{100 + c^2 - 81}{2 \times 10 \times c}$$

$$\Rightarrow \frac{1}{2} = \frac{19+c^2}{2 \times 10c} \Rightarrow c^2 - 10c + 19 = 0$$

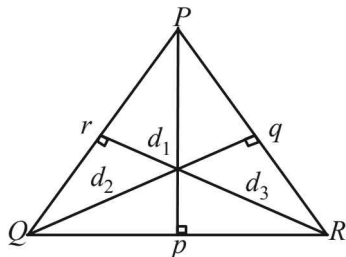
$$\Rightarrow c = \frac{10 \pm \sqrt{100-76}}{2} \Rightarrow c = 5 \pm \sqrt{6}$$

Given that $a = 10$, $b = 9$ are the longer sides

$$\therefore c = 5 \pm \sqrt{6} \text{ (both the values are less than 9 and 10),}$$

$$\therefore c = 5 + \sqrt{6} \text{ or } 5 - \sqrt{6} \text{ both are possible.}$$

3. (b) In $\triangle PQR$ let d_1, d_2, d_3 be the altitudes on QR, RP and PQ respectively.



$$\text{Then, Ar } (\triangle PQR) = \Delta = \frac{1}{2} p d_1 = \frac{1}{2} q d_2 = \frac{1}{2} r d_3$$

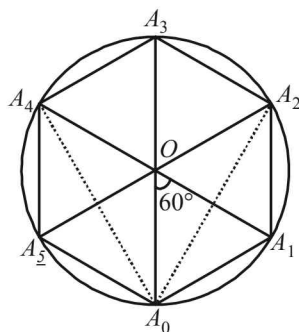
$$\Rightarrow d_1 = \frac{2\Delta}{p}, d_2 = \frac{2\Delta}{q}, d_3 = \frac{2\Delta}{r}$$

$$\Rightarrow d_1 = \frac{2\Delta}{k \sin P}, d_2 = \frac{2\Delta}{k \sin Q}, d_3 = \frac{2\Delta}{k \sin R}$$

[Using sine law]

$$\Rightarrow d_1, d_2, d_3 \text{ are in H.P. (As given that } \sin P, \sin Q, \sin R \text{ are in A.P.)}$$

4. (c) Given that $A_0 A_1 A_2 A_3 A_4 A_5$ is a regular hexagon inscribed in a circle of radius 1.



$$\angle A_0 O A_1 = \frac{360^\circ}{6} = 60^\circ$$

But in $\triangle O A_0 A_1$, $O A_0 = O A_1 = 1$

$$\therefore \angle O A_0 A_1 = \angle O A_1 A_0 = 60^\circ$$

$\therefore \triangle O A_0 A_1$ is an equilateral $\triangle$

$$\therefore A_0 A_1 = 1 = A_1 A_2 = A_2 A_3 = A_3 A_4 = A_4 A_5 = A_5 A_0$$

$$\angle A_0 A_1 A_2 = 60^\circ + 60^\circ = 120^\circ$$

In $\triangle A_0 A_1 A_2$, using cosine law we get

$$\cos 120^\circ = \frac{(A_0 A_1)^2 + (A_1 A_2)^2 - (A_0 A_2)^2}{2(A_0 A_1)(A_1 A_2)}$$

$$\Rightarrow -\frac{1}{2} = \frac{1+1-(A_0 A_2)^2}{2 \times 1 \times 1} \Rightarrow A_0 A_2 = \sqrt{3}$$

Thus by symmetry $A_0 A_4 = \sqrt{3}$

$$\therefore A_0 A_1 \cdot A_0 A_2 \cdot A_0 A_4 = 1 \cdot \sqrt{3} \cdot \sqrt{3} = 3$$

$\therefore$ (c) is the correct option.

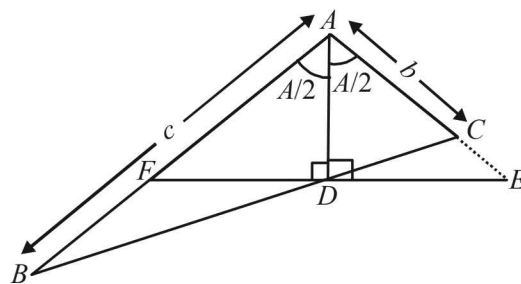
5. (a,b,c,d) By simple geometry in $\triangle AFE$, $AF = AE$

$\therefore \triangle AFE$ is an isosceles $\triangle$.

$$\text{Now Ar } (\triangle ABC) = \text{Ar } (\triangle ABD) + \text{Ar } (\triangle ADC)$$

$$\Rightarrow \frac{1}{2} bc \sin A = \frac{1}{2} c AD \sin \frac{A}{2} + \frac{1}{2} b AD \sin \frac{A}{2}$$

$$\Rightarrow AD = \frac{2bc \cos \frac{A}{2}}{b+c}$$



$$\text{Also } AD = AE \cos \frac{A}{2}$$

$$\Rightarrow AE = \frac{2bc}{b+c} = \text{HM of } b \text{ and } c.$$

$$\text{Again } EF = 2DE = 2 \cdot AD \tan \frac{A}{2} = \frac{4bc \sin \frac{A}{2}}{b+c}$$

6. (b) Given that $a = x^2 + x + 1$, $b = x^2 - 1$, $c = 2x + 1$ and $\angle C = \pi/6$

$$\text{Using } \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\text{we get } \cos \frac{\pi}{6} = \frac{(x^2 + x + 1)^2 + (x^2 - 1)^2 - (2x + 1)^2}{2(x^2 + x + 1)(x^2 - 1)}$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{(x^2 + 3x + 2)(x^2 - x) + (x^2 - 1)^2}{2(x^2 + x + 1)(x^2 - 1)}$$

$$\Rightarrow \sqrt{3}(x^2 + x + 1)(x + 1)(x - 1) = x(x - 1)(x + 1)(x + 2) + (x + 1)^2(x - 1)^2$$

$$\Rightarrow (x + 1)(x - 1) [\sqrt{3}(x^2 + x + 1) - x(x + 2) - (x + 1)(x - 1)] = 0$$

$$\Rightarrow (x + 1)(x - 1) [(\sqrt{3} - 2)x^2 + (\sqrt{3} - 2)x + (\sqrt{3} + 1)] = 0$$

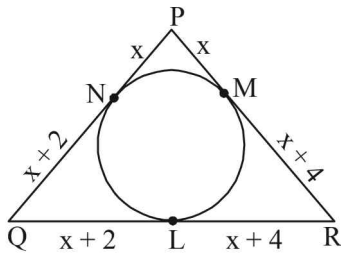
$$\Rightarrow x = -1, 1, (\sqrt{3} + 1), -(\sqrt{3} + 2)$$

Now for $x = -1$ and 1 , $b = 0$ which is not possible

Also for $x = -(\sqrt{3} + 2)$, $c = -4 - 2\sqrt{3} + 1 < 0$ which is not possible

$$\therefore x = \sqrt{3} + 1$$

7. (b, d) Let $PN = x$, $QL = x + 2$, $RM = x + 4$ where x is an even integer.



Then $PM = PN = x$, $QN = QL = x + 2$ and $RL = RM = x + 4$

So that $PQ = 2x + 2$, $QR = 2x + 6$, $PR = 2x + 4$

$$\text{Now } \cos P = \frac{1}{3} \Rightarrow \frac{PQ^2 + PR^2 - QR^2}{2PQ \cdot PR} = \frac{1}{3}$$

$$\Rightarrow \frac{(2x+2)^2 + (2x+4)^2 - (2x+6)^2}{2(2x+2)(2x+4)} = \frac{1}{3}$$

$$\Rightarrow 3[(x+1)^2 + (x+2)^2 - (x+3)^2] = 2(x+1)(x+2)$$

$$\Rightarrow 3(x^2 - 4) = 2(x+1)(x+2) \Rightarrow 3x - 6 = 2x + 2 \Rightarrow x = 8$$

$$\therefore PQ = 18, QR = 22, PR = 20$$

8. (a, c, d) $\frac{s-x}{4} = \frac{s-y}{3} = \frac{s-z}{2} = \frac{s-x+s-y+s-z}{4+3+2} = \frac{s}{9}$

$$\therefore x = \frac{5s}{9}, y = \frac{6s}{9}, z = \frac{7s}{9}$$

$$\text{Area of incircle} = \pi r^2 = \pi \frac{\Delta^2}{s^2} = \frac{8\pi}{3}$$

$$\Rightarrow \frac{s(s-x)(s-y)(s-z)}{s^2} = \frac{8}{3}$$

$$\Rightarrow \frac{4 \times 3 \times 2s^3}{9 \times 9 \times 9s} = \frac{8}{3} \text{ or } s = 9$$

$$\therefore x = 5, y = 6, z = 7$$

$$\begin{aligned} \text{(a) Area of } \triangle XYZ &= \sqrt{s(s-x)(s-y)(s-z)} \\ &= \sqrt{9 \times 4 \times 3 \times 2} = 6\sqrt{6} \end{aligned}$$

$$\text{(b) radius of circumcircle, } R = \frac{xyz}{4\Delta}$$

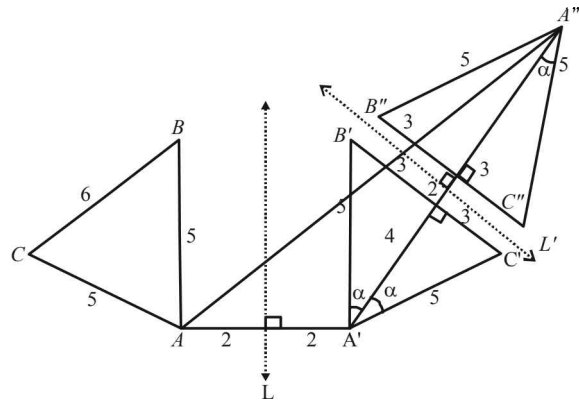
$$= \frac{5 \times 6 \times 7}{4 \times 6\sqrt{6}} = \frac{35\sqrt{6}}{24}$$

$$\begin{aligned} \text{(c) } r &= 4R \sin \frac{X}{2} \sin \frac{Y}{2} \sin \frac{Z}{2} \Rightarrow \sin \frac{X}{2} \sin \frac{Y}{2} \sin \frac{Z}{2} = \frac{r}{4R} \\ &= \frac{2\sqrt{2} \times 24}{\sqrt{3} \times 4 \times 35\sqrt{6}} = \frac{4}{35} \end{aligned}$$

$$\begin{aligned} \text{(d) } \sin^2 \frac{X+Y}{2} &= \cos^2 \frac{Z}{2} \\ &= \frac{s(s-z)}{xy} = \frac{9 \times 2}{5 \times 6} = \frac{3}{5} \end{aligned}$$

E. Subjective Problems

1. Let L be the line parallel to side AB of $\triangle ABC$, at a distance of 2 cm from AB , in which the first reflection $\triangle A'B'C'$ is obtained. Let L' be the second line parallel to $B'C'$, at a distance of 2 cm from $B'C'$, in which reflection of $\triangle A'B'C'$ is taken as $\triangle A''B''C''$. In figure, size of $\triangle A''B''C''$ is same to the size of $\triangle A'B'C'$.



From figure $AA' = 4$ cm and $A'A'' = 12$ cm. So to find AA'' it suffices to know $\angle AA'A''$, clearly

$$\angle AA'A'' = 90^\circ + \alpha \text{ where } \sin \alpha = \frac{3}{5}$$

$$\Rightarrow \cos \angle AA'A'' = \cos(90^\circ + \alpha) = -\sin \alpha = -\frac{3}{5} \text{ and hence}$$

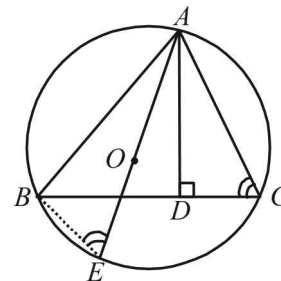
$$\begin{aligned} AA'' &= \sqrt{(AA')^2 + (A'A'')^2 - 2AA' \times A'A'' \cdot \cos(90^\circ + \alpha)} \\ &= \sqrt{16 + 144 + 96 \times \frac{3}{5}} \\ &= \sqrt{\frac{1088}{5}} = 8\sqrt{\frac{17}{5}} \text{ cm} \end{aligned}$$

2. (a) Inradius of the circle is given by

$$r = (s-b) \tan \frac{B}{2} = \left(\frac{a+b+c}{2} - b \right) \tan \frac{\pi}{4} = \frac{a+c-b}{2}$$

$$2r = a + c - b \Rightarrow \text{Diameter} = BC + AB - AC$$

- (b) Given a $\triangle ABC$ in which $AD \perp BC$, AE is diameter of circumcircle of $\triangle ABC$.



To prove :

$$AB \times AC = AE \times AD$$

Construction : Join BE

Proof : $\angle ABE = 90^\circ$

($\angle$ in a semi circle)

Now in $\triangle ABE$ and $\triangle ADC$

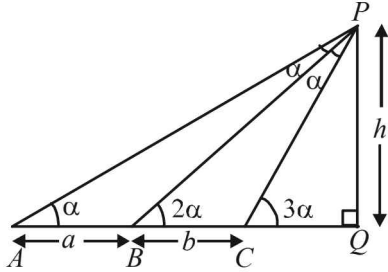
Properties of Triangle

$$\begin{aligned}\angle ABE &= \angle ADC && (\text{each } 90^\circ) \\ \angle AEB &= \angle ACD && (\angle\text{'s in the same segment}) \\ \therefore \Delta ABE &\sim \Delta ADC && (\text{by AA similarity})\end{aligned}$$

$$\Rightarrow \frac{AB}{AD} = \frac{AE}{AC}$$

$$\Rightarrow AB \times AC = AD \times AE \quad (\text{Proved})$$

3. By exterior angle theorem, in the adjacent fig.



$$\angle APB = \angle BPC = \alpha$$

$$\text{Also in } \Delta ABP, \angle BAP = \angle APB = \alpha$$

$$\Rightarrow AB = PB = a$$

Applying sine law in ΔPBC , we get

$$\Rightarrow \frac{a}{\sin(180^\circ - 3\alpha)} = \frac{b}{\sin \alpha} = \frac{PC}{\sin 2\alpha} \quad \dots(1)$$

$$\Rightarrow \frac{a}{3\sin \alpha - 4\sin^3 \alpha} = \frac{b}{\sin \alpha} = \frac{PC}{2\sin \alpha \cos \alpha}$$

$$\Rightarrow \frac{a}{3 - 4\sin^2 \alpha} = \frac{b}{1} = \frac{PC}{2\cos \alpha}$$

$$\Rightarrow 3 - 4\sin^2 \alpha = \frac{a}{b} \Rightarrow \sin^2 \alpha = \frac{3b - a}{4b}$$

$$\Rightarrow \cos^2 \alpha = \frac{b + a}{4b}$$

$$\Rightarrow \cos \alpha = \frac{1}{2} \sqrt{\frac{b + a}{b}}$$

$$\text{Also } PC = 2b \cos \alpha = \sqrt{b(a + b)}$$

Now in ΔPCQ

$$\sin 3\alpha = \frac{h}{PC} \Rightarrow h = PC \cdot \left(\frac{a \sin \alpha}{b} \right) \quad [\text{Using eqn. (1)}]$$

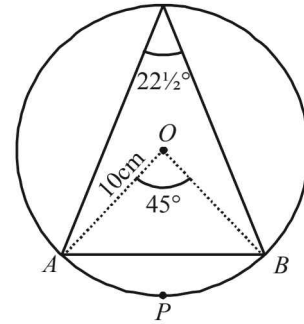
$$\Rightarrow h = \sqrt{b(a + b)} \cdot \frac{a}{b} \cdot \sqrt{\frac{3b - a}{4b}}$$

$$\Rightarrow h = \frac{a}{2b} \cdot \sqrt{(a + b)(3b - a)}$$

$$(b) \because \angle ACB = 22\frac{1}{2}^\circ$$

$$\therefore \angle AOB = 45^\circ$$

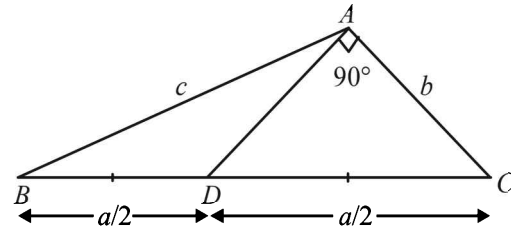
$$r = 10 \text{ cm}$$



Area of the segment APB = Area of the sector AOB – area of ΔAOB

$$\begin{aligned}&= \frac{1}{8} \pi r^2 - \frac{1}{2} \times 10 \times 10 \sin 45^\circ \quad (\text{Using, } \Delta = \frac{1}{2} bc \sin A) \\&= \frac{3.14 \times 100}{8} - \frac{50}{\sqrt{2}} = 3.91 \text{ sq. cm.}\end{aligned}$$

- 4.



$$\text{In } \Delta ACD, \cos C = \frac{b}{a/2} = \frac{2b}{a} \quad \dots(1)$$

$$\text{In } \Delta ABC, \cos C = \frac{a^2 + b^2 - c^2}{2ab} \quad \dots(2)$$

From (1) and (2),

$$\frac{2b}{a} = \frac{a^2 + b^2 - c^2}{2ab} \Rightarrow b^2 = \frac{1}{3} (a^2 - c^2) \quad \dots(3)$$

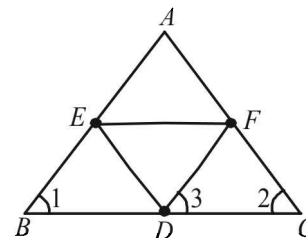
$$\text{Also } \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\therefore \cos A \cos C = \frac{b^2 + c^2 - a^2}{2bc} \times \frac{2b}{a} = \frac{b^2 + c^2 - a^2}{ac}$$

$$= \frac{\frac{1}{3}(a^2 - c^2) + (c^2 - a^2)}{ac} = \frac{2(c^2 - a^2)}{3ac}$$

5. Given that $AB = AC$

$$\therefore \angle 1 = \angle 2 \quad \dots(1)$$



But $AB \parallel DF$ (given) and BC is transversal

$$\therefore \angle 1 = \angle 3 \quad \dots(2)$$

From equations (1) and (2)

$$\angle 2 = \angle 3$$

$$\Rightarrow DF = CF \quad \dots(3)$$

Similarly we can prove $DE = BE$

$$\begin{aligned} \text{Now, } DF + FA + AE + ED &= CF + FA + AE + BE \\ &= AC + AB \quad [\text{using equation (3) and (4)}] \end{aligned}$$

6. (i) Let h be the height of tower PQ .

$$\text{In } \triangle APQ, \tan \theta = \frac{h}{AP} \Rightarrow AP = \frac{h}{\tan \theta}$$

Similarly in $\triangle$'s BPQ and CPQ we obtain

$$BP = \frac{h}{\tan \theta} = CP$$

$$\therefore AP = BP = CP$$

$\Rightarrow P$ is the circumcentre of

$$\triangle ABC \text{ with circum radius as } R = AP = \frac{abc}{4\Delta}$$

$$\therefore h = AP \tan \theta = \frac{abc \tan \theta}{4\Delta}$$

- (ii) Given $AP = AB \times n$

$$\Rightarrow \frac{AB}{AP} = \frac{1}{n} = \tan \theta$$

$\therefore$

$$\tan \theta = \frac{1}{n}$$

$$\text{Also } \tan(\theta - \beta) = \frac{AC}{AP} = \frac{1}{2} \frac{AB}{AP} = \frac{1}{2n}$$

$$\Rightarrow \frac{\tan \theta - \tan \beta}{1 + \tan \theta \tan \beta} = \frac{1}{2n} \Rightarrow \frac{\frac{1}{n} - \tan \beta}{1 + \frac{1}{n} \tan \beta} = \frac{1}{2n}$$

$$\Rightarrow 2n - 2n^2 \tan \beta = n + \tan \beta$$

$$\Rightarrow (2n^2 + 1) \tan \beta = n \Rightarrow \tan \beta = \frac{n}{2n^2 + 1}$$

7. As the angles A, B, C of $\triangle ABC$ are in AP

$$\therefore \text{ Let } A = x - d, B = x, C = x + d$$

$$\text{But } A + B + C = 180^\circ \quad (\angle \text{ Sum prop. of } \triangle)$$

$$\therefore x - d + x + x + d = 180^\circ$$

$$\Rightarrow 3x = 180^\circ \Rightarrow x = 60^\circ \therefore \angle B = 60^\circ$$

Now by sine law in $\triangle ABC$, we have

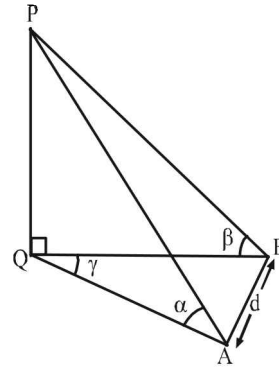
$$\frac{b}{\sin B} = \frac{c}{\sin C} \Rightarrow \frac{b}{c} = \frac{\sin B}{\sin C}$$

$$\Rightarrow \frac{\sqrt{3}}{\sqrt{2}} = \frac{\sin 60^\circ}{\sin C} \quad \left[\text{Using } b : c = \sqrt{3} : \sqrt{2} \right]$$

$$\Rightarrow \frac{\sqrt{3}}{\sqrt{2}} = \frac{\sqrt{3}}{2 \sin C} \Rightarrow \sin C = \frac{1}{\sqrt{2}} = \sin 45^\circ$$

$$\therefore \angle C = 45^\circ \Rightarrow \angle A = 180^\circ - (\angle B + \angle C) = 180^\circ - (60^\circ + 45^\circ) = 75^\circ$$

8. Let ht. of pole PQ be h .



$$\text{In } \triangle APQ, \tan \alpha = \frac{h}{AQ}$$

$$\Rightarrow AQ = \frac{h}{\tan \alpha} \quad \dots(1)$$

$$\text{In } \triangle BPQ, \tan \beta = \frac{h}{BQ} \Rightarrow BQ = \frac{h}{\tan \beta} \quad \dots(2)$$

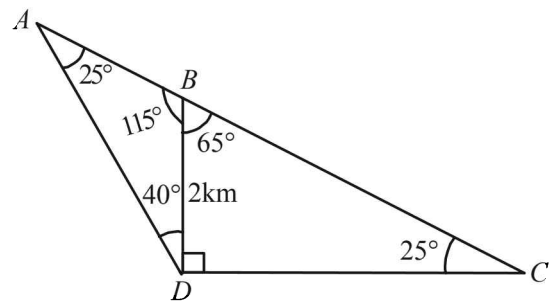
$$\text{In } \triangle ABQ, \cos \gamma = \frac{AQ^2 + BQ^2 - AB^2}{2AQ \cdot BQ}$$

$$\therefore \cos \gamma = \frac{h^2 \cot^2 \alpha + h^2 \cot^2 \beta - d^2}{2h^2 \cot \alpha \cot \beta}$$

$$\therefore -2h^2 \cot \alpha \cot \beta \cos \gamma + h^2 \cot^2 \alpha + h^2 \cot^2 \beta = d^2$$

$$\Rightarrow h = \frac{d}{\sqrt{\cot^2 \alpha + \cot^2 \beta - 2 \cot \alpha \cot \beta \cos \gamma}}$$

9. According to question figure is as follows



Here, $\angle BDC = 90^\circ, BD = 2 \text{ km}$

$$\angle BDA = 40^\circ \Rightarrow \angle ADC = 130^\circ,$$

$$\therefore \angle DAC = 180^\circ - (25^\circ + 130^\circ) = 25^\circ$$

From the figure, in $\triangle ABD$, using sine law

$$\frac{AD}{\sin 115^\circ} = \frac{BD}{\sin 25^\circ} \Rightarrow$$

$$AD = \frac{2 \sin(90^\circ + 25^\circ)}{\sin 25^\circ} = \frac{2 \cos 25^\circ}{\sin 25^\circ}$$

$$\Rightarrow AD = 2 \cot 25^\circ$$

$$= 2 \sqrt{\frac{1}{\sin^2 25^\circ} - 1} = 2 \sqrt{\frac{1}{(0.423)^2} - 1} = 4.28 \text{ km}$$

Properties of Triangle

10. KEY CONCEPT:

Ex-radii of a $\triangle ABC$ are $r_1 = \frac{\Delta}{s-a}$, $r_2 = \frac{\Delta}{s-b}$, $r_3 = \frac{\Delta}{s-c}$. As r_1, r_2, r_3 are in HP. $\therefore \frac{1}{r_1}, \frac{1}{r_2}, \frac{1}{r_3}$ are in AP

$$\Rightarrow \frac{s-a}{\Delta}, \frac{s-b}{\Delta}, \frac{s-c}{\Delta} \text{ are in AP}$$

$$\Rightarrow s-a, s-b, s-c \text{ are in AP}$$

$$\Rightarrow -a, -b, -c \text{ are in AP}$$

$$\Rightarrow a, b, c \text{ are in A.P.}$$

11. Given that $\cos A + \cos B + \cos C = \frac{3}{2}$ in $\triangle ABC$

$$\Rightarrow 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2} = \frac{3}{2} - \cos C$$

$$\Rightarrow 2 \sin \frac{C}{2} \cos \frac{A-B}{2} = \frac{3-2\cos C}{2}$$

$$\Rightarrow 2 \sin \frac{C}{2} \cos \frac{A-B}{2} = \frac{3-2(1-2\sin^2 C/2)}{2}$$

$$\Rightarrow \cos \left(\frac{A-B}{2} \right) = \frac{1+4\sin^2 C/2}{4\sin C/2}$$

$$\Rightarrow \cos \left(\frac{A-B}{2} \right) = \frac{1+4\sin^2 C/2 - 4\sin C/2 + 4\sin C/2}{4\sin C/2}$$

$$\Rightarrow \cos \left(\frac{A-B}{2} \right) = \frac{(1-2\sin C/2)^2}{4\sin C/2} + 1$$

Which is possible only when

$$1-2\sin C/2 = 0 \Rightarrow \sin C/2 = 1/2$$

$$\Rightarrow C/2 = 30^\circ \Rightarrow C = 60^\circ$$

$$\text{Also then } \cos \frac{A-B}{2} = 1 \Rightarrow \frac{A-B}{2} = 0 \Rightarrow A-B=0 \quad \dots(1)$$

$$\text{and } A+B = 180^\circ - 60^\circ = 120^\circ \Rightarrow A+B=120^\circ \quad \dots(2)$$

From (1) and (2) $A=B=60^\circ$

Thus we get $A=B=C=60^\circ \therefore \triangle ABC$ is an equilateral $\triangle$.

12. Given that, in $\triangle ABC$,

$$\frac{b+c}{11} = \frac{c+a}{12} = \frac{a+b}{13}$$

where a, b, c are the lengths of sides BC, CA and AB respectively.

$$\text{Let } \frac{b+c}{11} = \frac{c+a}{12} = \frac{a+b}{13} = k$$

$$\Rightarrow b+c=11k \quad \dots(1)$$

$$c+a=12k \quad \dots(2)$$

$$a+b=13k \quad \dots(3)$$

Adding the above three eqs. we get

$$\Rightarrow 2(a+b+c) = 36k$$

$$\Rightarrow a+b+c = 18k \quad \dots(4)$$

Solving each of (1), (2) and (3) with (4), we get

$$\text{Now, } \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$= \frac{36k^2 + 25k^2 - 49k^2}{2 \times 6k \times 5k} = \frac{12k^2}{60k^2} = \frac{1}{5}$$

$$\cos B = \frac{c^2 + a^2 - b^2}{2ca} = \frac{25k^2 + 49k^2 - 36k^2}{2 \times 5k \times 7k}$$

$$= \frac{38k^2}{70k^2} = \frac{19}{35}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{49k^2 + 36k^2 - 25k^2}{2 \times 7k \times 6k}$$

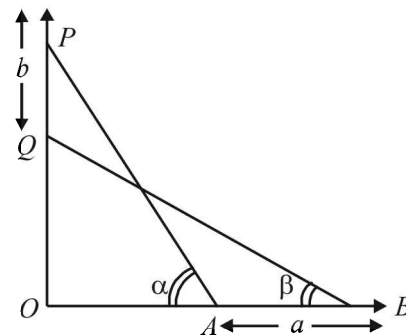
$$= \frac{60k^2}{84k^2} = \frac{5}{7}$$

$$\therefore \frac{\cos A}{1/5} = \frac{\cos B}{19/35} = \frac{\cos C}{5/7}$$

$$\Rightarrow \frac{\cos A}{7/35} = \frac{\cos B}{19/35} = \frac{\cos C}{25/35}$$

$$\Rightarrow \frac{\cos A}{7} = \frac{\cos B}{19} = \frac{\cos C}{25}$$

13. Let ℓ be the length of the ladder, then



$$\text{In } \triangle OQB, \cos \beta = \frac{OB}{BQ}$$

$$\Rightarrow OB = \ell \cos \beta \quad \dots(1)$$

$$\text{Similarly in } \triangle OPA, \cos \alpha = \frac{OA}{PA}$$

$$\Rightarrow OA = \ell \cos \alpha \quad \dots(2)$$

$$\text{Now } a = OB - OA = \ell (\cos \beta - \cos \alpha) \quad \dots(3)$$

Also from $\triangle OAP$, $OP = \ell \sin \alpha$

and in $\triangle OQB$, $OQ = \ell \sin \beta$

$$\therefore b = OP - OQ = \ell (\sin \alpha - \sin \beta) \quad \dots(4)$$

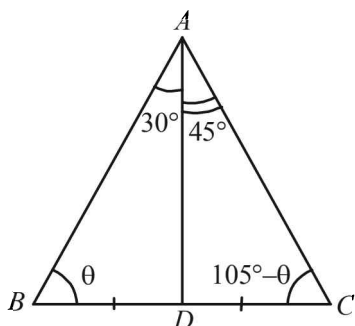
Dividing eq. (3) by (4) we get

$$\frac{a}{b} = \frac{\cos \beta - \cos \alpha}{\sin \alpha - \sin \beta}$$

$$\Rightarrow \frac{a}{b} = \frac{2 \sin \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\alpha - \beta}{2} \right)}{2 \cos \left(\frac{\alpha + \beta}{2} \right) - \sin \left(\frac{\alpha - \beta}{2} \right)} \Rightarrow \frac{a}{b} = \tan \left(\frac{\alpha + \beta}{2} \right)$$

Thus, $a = b \tan \left(\frac{\alpha + \beta}{2} \right)$ is proved.

14. Let AD be the median in $\triangle ABC$.
Let $\angle B = \theta$ then $\angle C = 105^\circ - \theta$
In $\triangle ABD$, using sine law, we get



$$\frac{BD}{\sin 30^\circ} = \frac{AD}{\sin \theta} \Rightarrow BD = \frac{AD}{2 \sin \theta}$$

In $\triangle ACD$, using sine law, we get

$$\frac{DC}{\sin 45^\circ} = \frac{AD}{\sin(105^\circ - \theta)} \Rightarrow DC = \frac{AD}{\sqrt{2} \sin(105^\circ - \theta)}$$

As $BD = DC$

$$\Rightarrow \frac{AD}{2 \sin \theta} = \frac{AD}{\sqrt{2} \sin(105^\circ - \theta)}$$

$$\Rightarrow \sin(90^\circ + 15^\circ - \theta) = \sqrt{2} \sin \theta$$

$$\Rightarrow \cos 15^\circ \cos \theta + \sin 15^\circ \sin \theta = \sqrt{2} \sin \theta$$

$$\Rightarrow \cot \theta = \frac{\sqrt{2} - \sin 15^\circ}{\cos 15^\circ} = \frac{5 - \sqrt{3}}{\sqrt{3} + 1} = 3\sqrt{3} - 4$$

$$\Rightarrow \operatorname{cosec} \theta = \sqrt{1 + \cot^2 \theta} = \sqrt{1 + 27 + 16 - 24\sqrt{3}}$$

$$\Rightarrow \operatorname{cosec} \theta = 2\sqrt{11 - 6\sqrt{3}}$$

$$\therefore BD = \frac{AD}{2 \sin \theta} = \frac{1}{\sqrt{11 - 6\sqrt{3}}} \times \frac{2\sqrt{11 - 6\sqrt{3}}}{2} = 1$$

$$\therefore BC = 2BD = 2 \text{ units}$$

15. We are given that in $\triangle ABC$ $\cos A \cos B + \sin A \sin B \sin C = 1$

$$\Rightarrow \sin A \sin B \sin C = 1 - \cos A \cos B$$

$$\Rightarrow \sin C = \frac{1 - \cos A \cos B}{\sin A \sin B}$$

$$\Rightarrow \frac{1 - \cos A \cos B}{\sin A \sin B} \leq 1 \quad [\because \sin C \leq 1]$$

$$\Rightarrow 1 - \cos A \cos B \leq \sin A \sin B$$

$$\Rightarrow 1 \leq \cos A \cos B + \sin A \sin B$$

$$\Rightarrow 1 \leq \cos(A - B) \Rightarrow 1 \leq \cos(A - B)$$

But we know $\cos(A - B) \leq 1$

$$\therefore \text{We must have } \cos(A - B) = 1$$

$$\Rightarrow A - B = 0$$

$$\Rightarrow A = B$$

$$\therefore \cos A \cos A + \sin A \sin A \sin C = 1 \quad [\text{For } A = B]$$

$$\Rightarrow \cos^2 A + \sin^2 A \sin C = 1$$

$$\Rightarrow \sin^2 A \sin C = 1 - \cos^2 A$$

$$\Rightarrow \sin^2 A \sin C = \sin^2 A \Rightarrow \sin^2 A (\sin C - 1) = 0$$

$$\Rightarrow \sin A = 0 \text{ or } \sin C = 1$$

The only possibility is $\sin C = 1 \Rightarrow C = \pi/2$

$$\therefore A + B = \pi/2$$

But $A = B \Rightarrow A = B = \pi/4$

$\therefore$ By Sine law in $\triangle ABC$,

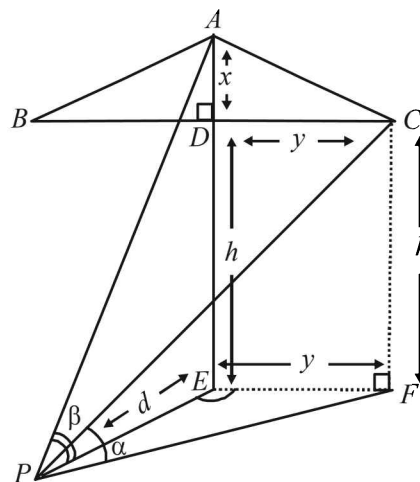
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$\Rightarrow \frac{a}{\sin 45^\circ} = \frac{b}{\sin 45^\circ} = \frac{c}{\sin 90^\circ} \Rightarrow \frac{a}{1/\sqrt{2}} = \frac{b}{1/\sqrt{2}} = \frac{c}{1}$$

$$\Rightarrow \frac{a}{1} = \frac{b}{1} = \frac{1}{\sqrt{2}} \Rightarrow a : b : c = 1 : 1 : \sqrt{2}$$

Hence proved the result.

16. Let ABC be the isosceles triangular sign board with BC horizontal. DE be the pole of height h . Let the man be standing at P such that $PE = d$



Also $\angle APE = \beta$

$\angle CPF = \alpha$

Let $AD = x$ be altitude of $\triangle ABC$.

As $\triangle ABC$ is isosceles with $AB = AC$

$\therefore D$ is mid pt of BC .

Hence $BC = 2y$.

Now in $\triangle APE$,

$$\tan \beta = \frac{h+x}{d} \Rightarrow x = d \tan \beta - h \quad \dots(1)$$

$$\text{In } \triangle CPF, \tan \alpha = \frac{h}{PF} \Rightarrow \tan \alpha = \frac{h}{\sqrt{d^2 + y^2}}$$

$$\Rightarrow y^2 + d^2 = h^2 \cot^2 \alpha$$

$$\Rightarrow y = \sqrt{h^2 \cot^2 \alpha - d^2} \quad \dots(2)$$

$$\text{Now area of } \triangle ABC = \frac{1}{2} \times BC \times AD$$

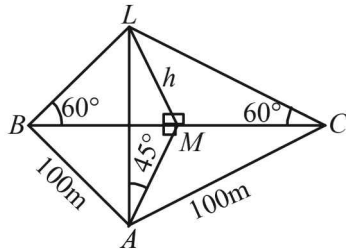
$$= \frac{1}{2} \times 2y \times x = xy$$

$$= (d \tan \beta - h) \sqrt{h^2 \cot^2 \alpha - d^2}$$

[Using (1) and (2)]

Properties of Triangle

17. Let ABC be the triangular region with $AB = AC = 100$ m. Let M be the mid pt of BC at which tower LM stands. As $\triangle ABC$ is isosceles and M is mid pt. of BC
 $\therefore AM \perp BC$.
 Let $LM = h$ be the ht. of tower.



In $\triangle ALM$, $\tan 45^\circ = \frac{LM}{MA} \Rightarrow LM = MA$
 $\therefore MA = h$

Also in $\triangle BLM$, $\tan 60^\circ = \frac{LM}{BM}$

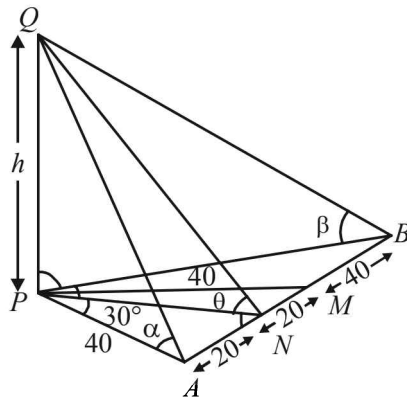
$\Rightarrow \sqrt{3} = \frac{h}{BM} \Rightarrow BM = \frac{h}{\sqrt{3}}$

Now in $rt \triangle AMB$, we have,
 $AB^2 = AM^2 + BM^2$

$\Rightarrow (100)^2 = h^2 + \frac{h^2}{3} \Rightarrow \frac{4h^2}{3} = 10000$

$\Rightarrow h = 50\sqrt{3}$ m.

18. Let $PQ = h$
 As A and B are located to the south and east of P respectively,
 $\therefore \angle APB = 90^\circ$. M is mid pt of AB . PAM is an equilateral $\triangle$



$\therefore \angle APM = 60^\circ$:

Also $PN \perp AB$, therefore $AN = NM = 20$ m

$\Rightarrow AP = 40$ m

Let angles of elevation of top of the tower from A , N and B be α , θ and β respectively. At Q , $\tan \theta = 2$

In $\triangle PQN$ $\tan \theta = \frac{PQ}{PN}$

$\Rightarrow 2 = \frac{h}{PN} \Rightarrow PN = h/2$ (1)

Also in $\triangle APM$, $\angle APM = 60^\circ$ (being equilateral $\triangle$) and PN is altitude $\therefore \angle APN = 30^\circ$ (as in equilateral $\triangle$ altitude bisects the vertical angle).

$\therefore$ In $\triangle APN$ $\tan \angle APN = \frac{AN}{PN}$

$\Rightarrow \tan 30^\circ = \frac{20}{h/2}$ [Using eq. (1)]

$\Rightarrow \frac{h}{2\sqrt{3}} = 20 \Rightarrow h = 40\sqrt{3}$ m.

In $\triangle APQ$ $\tan \alpha = \frac{h}{AP} \Rightarrow \tan \alpha = \frac{40\sqrt{3}}{40} = \sqrt{3}$

$\Rightarrow \alpha = 60^\circ$ Also in $\triangle ABQ$ $\tan \beta = h/PB$ But in $rt \triangle PNB$

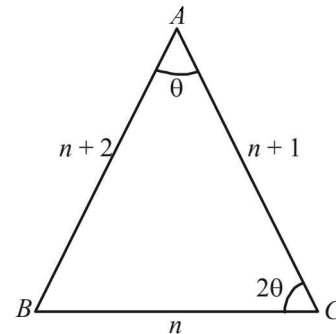
$PB = \sqrt{PN^2 + NB^2} = \sqrt{(20\sqrt{3})^2 + (60)^2}$

$\therefore PB = \sqrt{1200 + 3600} = \sqrt{4800} = 40\sqrt{3}$

$\therefore \tan \beta = \frac{40\sqrt{3}}{40\sqrt{3}} \Rightarrow \tan \beta = 1 \Rightarrow \beta = 45^\circ$

Thus $h = 40\sqrt{3}$ m ; $\angle$'s of elevation are $60^\circ, 45^\circ$.

19. Let the sides of $\triangle$ be $n, n+1, n+2$ where $n \in \mathbb{N}$.
 Let $a = n, b = n+1, c = n+2$



Let the smallest angle $\angle A = \theta$ then the greatest $\angle C = 2\theta$
 In $\triangle ABC$ by applying Sine Law we get,

$\frac{\sin \theta}{n} = \frac{\sin 2\theta}{n+2}$

$\Rightarrow \frac{\sin \theta}{n} = \frac{2 \sin \theta \cos \theta}{n+2} \Rightarrow \frac{1}{n} = \frac{2 \cos \theta}{n+2}$ (as $\sin \theta \neq 0$)

$\Rightarrow \cos \theta = \frac{n+2}{2n}$ (1)

In $\triangle ABC$ by Cosine Law, we get

$\cos \theta = \frac{(n+1)^2 + (n+2)^2 - n^2}{2(n+1)(n+2)}$ (2)

Comparing the values of $\cos \theta$ from (1) and (2), we get

$\frac{(n+1)^2 + (n+2)^2 - n^2}{2(n+1)(n+2)} = \frac{n+2}{2n}$

$\Rightarrow (n+2)^2 (n+1) = n(n+2)^2 + n(n+1)^2 - n^3$

$\Rightarrow n(n+2)^2 + (n+2)^2 = n(n+2)^2 + n(n+1)^2 - n^3$

$\Rightarrow n^2 + 4n + 4 = n^3 + 2n^2 + n - n^3$

$\Rightarrow n^2 - 3n - 4 = 0 \Rightarrow (n+1)(n-4) = 0$

$\Rightarrow n = 4$ (as $n \neq -1$)

$\therefore$ Sides of $\triangle$ are 4, 4+1, 4+2, i.e. 4, 5, 6.

20. Given that, In $\triangle ABC$, base = a

$$\text{and } \frac{c}{b} = r$$

To find altitude, h .

We have, in $\triangle ABD$,

$$h = c \sin B = \frac{ca \sin B}{a}$$

$$= \frac{c k \sin A \sin B}{k \sin A} = \frac{c k \sin A \sin B}{\sin(B+C)}$$

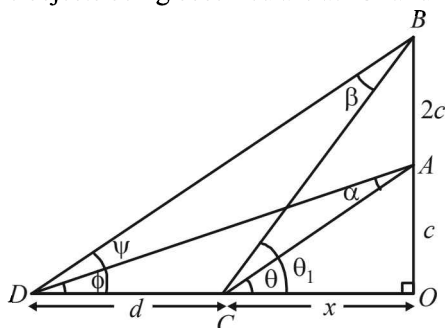
$$= \frac{c \sin A \sin B \sin(B-C)}{\sin(B+C) \sin(B-C)} = \frac{c \sin A \sin B \sin(B-C)}{\sin^2 B - \sin^2 C}$$

$$= \frac{c \cdot \frac{a}{k} \cdot \frac{b}{k} \cdot \sin(B-C)}{\frac{b^2}{k^2} - \frac{c^2}{k^2}} = \frac{abc \sin(B-C)}{b^2 - c^2}$$

$$= \frac{a \left(\frac{c}{b} \right) \sin(B-C)}{1 - \left(\frac{c}{b} \right)^2} = \frac{ar \sin(B-C)}{1 - r^2} \leq \frac{ar}{1 - r^2}$$

$[\because \sin(B-C) \leq 1] \therefore h \leq \frac{ar}{1-r^2}$ Hence Proved.

21. Let the man initially be standing at 'A' and 'B' be the position after walking a distance 'c', so total distance becomes $2c$ and the objects being observed are at 'C' and 'D'.



Now we have $OA = c$, $AB = 2c$

Let $CO = x$ and $CD = d$

Let $\angle CAD = \alpha$ and $\angle CBD = \beta$

$\angle ACO = \theta$ and $\angle ADC = \phi$

$\angle BCD = \psi$ and $\angle BCO = \theta_1$

$$\text{In } \triangle ACO, \quad \tan \theta = \frac{AO}{CO} \Rightarrow \tan \theta = \frac{c}{x} \quad \dots(1)$$

$$\text{In } \triangle ADO, \quad \tan \phi = \frac{c}{x+d} \quad \dots(2)$$

Now, $\theta = \alpha + \phi$ (Using ext. $\angle$ thm.)

$$\Rightarrow \alpha = \theta - \phi \Rightarrow \tan \alpha = \tan(\theta - \phi) \Rightarrow \frac{\tan \theta - \tan \phi}{1 + \tan \theta \tan \phi}$$

$$= \frac{\frac{c}{x} - \frac{c}{x+d}}{1 + \frac{c}{x} \cdot \frac{c}{x+d}} \quad (\text{Using equations (1) and (2)})$$

$$\Rightarrow \tan \alpha = \frac{cx + cd - cx}{x^2 + dx + c^2}$$

$$\Rightarrow x^2 + c^2 + xd = cd \cot \alpha \quad \dots(3)$$

Again in $\triangle ADO$

$$\tan \psi = \frac{3c}{x+d} \quad \dots(4)$$

$$\tan \theta_1 = \frac{3c}{x} \quad \dots(5)$$

But $\theta_1 = \psi + \beta$ (by ext. $\angle$ thm)

$$\Rightarrow \beta = \theta_1 - \psi$$

$$\Rightarrow \tan \beta = \tan(\theta_1 - \psi) = \frac{\tan \theta_1 - \tan \psi}{1 + \tan \theta_1 \tan \psi}$$

$$\Rightarrow \tan \beta = \frac{\frac{3c}{x} - \frac{3c}{x+d}}{1 + \frac{3c}{x} \cdot \frac{3c}{x+d}} \quad [\text{Using (4) and (5)}]$$

$$\Rightarrow \tan \beta = \frac{3cd}{x^2 + xd + 9c^2}$$

$$\Rightarrow x^2 + xd + 9c^2 = 3cd \cot \beta \quad \dots(6)$$

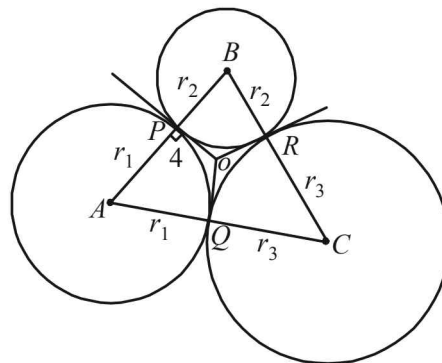
From (3) and (6), we get

$$8c^2 = 3cd \cot \beta - cd \cot \alpha$$

$$\Rightarrow d = \frac{8c}{3 \cot \beta - \cot \alpha} \quad \text{Hence Proved.}$$

22. Let us consider three circles with centres at A , B and C and with radii r_1 , r_2 and r_3 respectively which touch each other externally at P , Q and R . Let the common tangents at P , Q and R meet each other at O . Then $OP = OQ = OR = 4$ (given) (lengths of tangents from a pt to a circle are equal).

Also $OP \perp AB$, $OQ \perp AC$, $OR \perp BC$.



$\Rightarrow O$ is the incentre of the $\triangle ABC$

$$\text{Thus for } \triangle ABC, \quad s = \frac{(r_1 + r_2) + (r_2 + r_3) + (r_3 + r_1)}{2}$$

$$\text{i.e. } s = (r_1 + r_2 + r_3)$$

$$\therefore \Delta = \sqrt{(r_1 + r_2 + r_3) \cdot r_1 \cdot r_2 \cdot r_3} \quad (\text{Heron's formula})$$

Properties of Triangle

Now $r = \frac{\Delta}{s}$

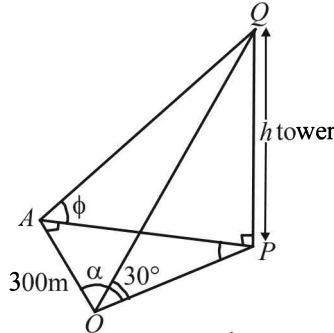
NOTE THIS STEP:

$$\Rightarrow 4 = \frac{\sqrt{(r_1 + r_2 + r_3) r_1 r_2 r_3}}{r_1 + r_2 + r_3} \Rightarrow 4 = \frac{\sqrt{r_1 r_2 r_3}}{\sqrt{r_1 + r_2 + r_3}}$$

$$\Rightarrow \frac{r_1 \cdot r_2 \cdot r_3}{r_1 + r_2 + r_3} = \frac{16}{1} \Rightarrow r_1 \cdot r_2 \cdot r_3 : r_1 + r_2 + r_3 = 16 : 1$$

23. Let PQ be the tower of height h . A is in the north of O and P is towards east of A .

$$\therefore \angle OAP = 90^\circ; \angle QOP = 30^\circ; \angle QAP = \phi$$



$$\angle AOP = \alpha \text{ s.t. } \tan \alpha = \frac{1}{\sqrt{2}}$$

$$\text{Now in } \triangle OPQ, \tan 30^\circ = \frac{h}{OP} \Rightarrow OP = h\sqrt{3} \quad \dots(1)$$

$$\text{In } \triangle APQ, \tan \phi = \frac{h}{AP} \Rightarrow AP = h \cot \phi \quad \dots(2)$$

$$\text{Given that, } \tan \alpha = \frac{1}{\sqrt{2}} \Rightarrow \sin \alpha = \frac{\tan \alpha}{\sqrt{1 + \tan^2 \alpha}} = \frac{1}{3}$$

$$\text{Now in } \triangle AOP, \sin \alpha = \frac{AP}{OP} \Rightarrow \frac{1}{3} = \frac{h \cot \phi}{h\sqrt{3}}$$

[Using (1) and (2)]

$$\Rightarrow \cot \phi = 1 \Rightarrow \phi = 45^\circ.$$

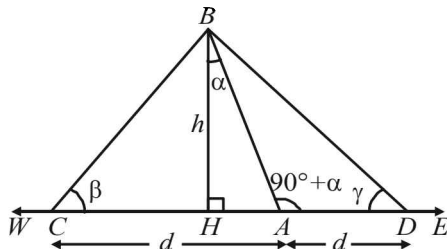
Again in $\triangle OAP$, using Pythagoras thm, we get

$$OP^2 = OA^2 + AP^2$$

$$\Rightarrow 3h^2 = 90000 + h^2 \cot^2 45^\circ \Rightarrow h = 150\sqrt{2} \text{ m}$$

24. Let AB be the tower leaning towards west making an angle α with vertical

At C , $\angle$ of elevation of B is β and at D the



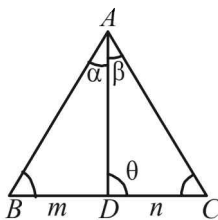
$\angle$ of elevation of B is γ

$$CA = AD = d$$

KEY CONCEPT:

$m : n$ theorem: In $\triangle ABC$ where point D divides BC in the ratio $m : n$. and $\angle ADC = \theta$

$$(i) (m + n) \cot \theta = n \cot B - m \cot C$$



$$(ii) (m + n) \cot \theta = m \cot \alpha - n \cot \beta$$

In $\triangle BCD$, A divides CD in the ratio $1 : 1$ where base $\angle$'s are β and γ and $\angle BAD = 90^\circ + \alpha$

$\therefore$ By applying $m : n$ theorem we get

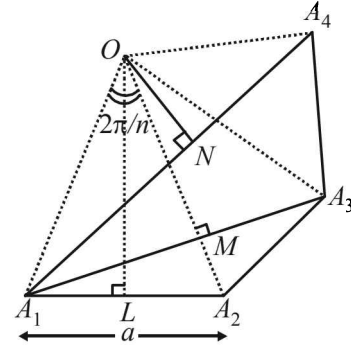
$$(1 + 1) \cot (90^\circ + \alpha) = 1 \cdot \cot \beta - 1 \cdot \cot \gamma$$

$$\Rightarrow -2 \tan \alpha = \cot \beta - \cot \gamma$$

$$\Rightarrow 2 \tan \alpha = \cot \gamma - \cot \beta$$

Hence Proved.

25. Let a be the side of n sided regular polygon $A_1 A_2 A_3 A_4 \dots A_n$



$$\therefore \angle \text{Subtended by each side at centre} = \frac{2\pi}{n}$$

Let $OL \perp A_1 A_2$

$$\text{Then } \angle OLA_1 = 90^\circ, \angle A_1 OL = \pi/n \quad (\because OA_1 = OA_2)$$

$$\therefore \text{In } \triangle OA_1 L, \sin A_1 OL = \frac{A_1 L}{OA_1} \Rightarrow \sin \pi/n = \frac{a/2}{OA_1}$$

$$\Rightarrow OA_1 = \frac{a/2}{\sin \pi/n} \quad \dots(1)$$

Again by geometry it can be proved that $OM \perp A_1 A_3$

$$\text{In } \triangle OA_1 M, \sin 2\pi/n = \frac{A_1 M}{OA_1} \Rightarrow A_1 M = OA_1 \sin 2\pi/n$$

$$\Rightarrow A_1 A_3 = \frac{2a \sin 2\pi/n}{2 \sin \pi/n} \quad [\text{Using eq}^n (1)]$$

Also if $ON \perp A_1 A_4$, then ON bisects angle

$$\angle A_1 OA_4 = 3(2\pi/n)$$

$$\therefore \angle A_1 ON = 3\pi/n$$

$$\therefore \text{In } \triangle OA_1 N, \sin 3\pi/2 = \frac{A_1 N}{OA_1}$$

$$\Rightarrow A_1 N = OA_1 \sin 3\pi/n$$

$$\Rightarrow A_1 A_4 = 2A_1 N = \frac{2a \sin 3\pi/n}{2 \sin \pi/n}$$

But given that

$$\frac{1}{A_1 A_2} = \frac{1}{A_1 A_3} + \frac{1}{A_1 A_4}$$

$$\Rightarrow \frac{1}{a} = \frac{1}{\frac{a \sin 2\pi/n}{\sin \pi/n}} + \frac{1}{\frac{a \sin 3\pi/n}{\sin \pi/n}}$$

$$\Rightarrow \sin 3\pi/n \sin 2\pi/n = (\sin 3\pi/n + \sin 2\pi/n) \sin \pi/n$$

$$\begin{aligned}
 &\Rightarrow 2 \sin 3\pi/n \sin 2\pi/n \\
 &\quad = 2 \sin 3\pi/n \sin \pi/n + 2 \sin 2\pi/n \sin \pi/n \\
 &\Rightarrow \cos \pi/n - \cos 5\pi/n \\
 &\quad = \cos 2\pi/n - \cos 4\pi/n + \cos \pi/n - \cos 3\pi/n \\
 &\Rightarrow \cos 2\pi/n + \cos 5\pi/n = \cos 4\pi/n + \cos 3\pi/n \\
 &\Rightarrow 2 \cos 7\pi/2n \cos 3\pi/2n = 2 \cos 7\pi/2n \cos \pi/2n \\
 &\Rightarrow \cos 7\pi/2n (\cos 3\pi/2n - \cos \pi/2n) = 0 \\
 &\Rightarrow \cos 7\pi/2n \cdot 2 \sin 2\pi/n \sin \pi/n = 0 \\
 &\Rightarrow \cos 7\pi/2n = 0 \\
 &\text{or } \sin 2\pi/n = 0 \text{ or } \sin \pi/n = 0 \\
 &\Rightarrow \frac{7\pi}{2n} = (2k+1)\pi/2 \text{ or } \frac{2\pi}{n} = k\pi \text{ or } \pi/n = p\pi \\
 &\Rightarrow n = \frac{7}{2k+1} \text{ or } n = 2/k \text{ or } n = 1/k
 \end{aligned}$$

But n should be a +ve integer being no. of sides and $n \geq 4$ (four vertices being considered in the question)

$$\therefore \text{ the only possibility is } n = \frac{7}{2k+1} \text{ for } k=0$$

$$\therefore n=7$$

26. (I) a, b, c and Δ are rational.

$$\Rightarrow s = \frac{a+b+c}{2} \text{ is also rational}$$

$$\Rightarrow \tan B/2 = \sqrt{\frac{(s-a)(s-c)}{s(s-b)}} = \frac{\Delta}{s(s-b)} \text{ is also rational}$$

$$\text{and } \tan C/2 = \sqrt{\frac{(s-a)(s-b)}{s(s-c)}} = \frac{\Delta}{s(s-c)} \text{ is also rational}$$

Hence (I) $\Rightarrow$ (II).

(II) $a, \tan B/2, \tan C/2$ are rational.

$$\Rightarrow \sin B = \frac{2 \tan B/2}{1 + \tan^2 B/2}$$

$$\text{and } \sin C = \frac{2 \tan C/2}{1 + \tan^2 C/2} \text{ are rationals.}$$

$$\text{and } \tan A/2 = \tan \left[90^\circ - \left(\frac{B}{2} + \frac{C}{2} \right) \right] = \cot \left(\frac{B}{2} + \frac{C}{2} \right)$$

$$= \frac{1}{\tan \left(\frac{B}{2} + \frac{C}{2} \right)} = \frac{1 - \tan B/2 \tan C/2}{\tan B/2 + \tan C/2} \text{ is rational}$$

$$\therefore \sin A = \frac{2 \tan A/2}{1 + \tan^2 A/2} \text{ is rational.}$$

Hence (II) $\Rightarrow$ (III)

(III) $a, \sin A, \sin B, \sin C$ are rational.

$$\text{But } \frac{a}{\sin A} = 2R \Rightarrow R \text{ is rational}$$

$$\therefore b = 2R \sin B, c = 2R \sin C \text{ are rationals.}$$

$$\therefore \Delta = \frac{1}{2} bc \sin A \text{ is rational}$$

Hence (III) $\Rightarrow$ (I).

27. Let A, B and C be the projections of the pts.

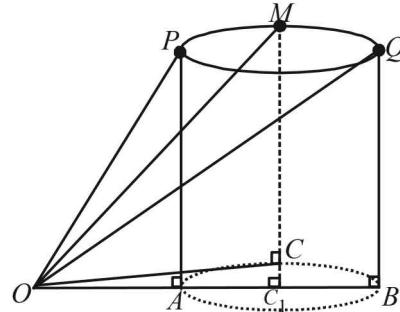
P, Q and M on the ground.

ATQ, $\angle POA = 60^\circ, \angle QOB = 30^\circ, \angle MOC = \theta$

Let h be the ht of circle from ground, then

$$AP = CM = BQ = h$$

Let $OA = x$ and $AB = d$ (diameter of the projection of the circle on ground with C_1 as centre).



$$\text{Now in } \triangle POA, \tan 60^\circ = \frac{h}{x} \Rightarrow x = \frac{h}{\sqrt{3}} \quad \dots(1)$$

$$\text{In } \triangle QBO, \tan 30^\circ = \frac{h}{x+d} \Rightarrow x+d = h\sqrt{3}$$

$$\Rightarrow d = h\sqrt{3} - \frac{h}{\sqrt{3}} = \frac{2h}{\sqrt{3}} \quad \dots(2)$$

$$\text{In } \triangle OMC, \tan \theta = \frac{h}{OC}$$

$$\Rightarrow \tan^2 \theta = \frac{h^2}{OC^2} = \frac{h^2}{OC_1^2 + C_1C^2} = \frac{h^2}{\left(x + \frac{d}{2}\right)^2 + \left(\frac{d}{2}\right)^2}$$

$$= \frac{h^2}{\left(\frac{h}{\sqrt{3}} + \frac{h}{\sqrt{3}}\right)^2 + \left(\frac{h}{\sqrt{3}}\right)^2} \quad [\text{Using (1) and (2)}]$$

$$= \frac{h^2}{\frac{4h^2}{3} + \frac{h^2}{3}} = \frac{3}{5}$$

28. Let ABC is an equilateral Δ then

$$A = B = C = 60^\circ$$

$$\Rightarrow \tan A + \tan B + \tan C = 3\sqrt{3}$$

Conversely, suppose

$$\tan A + \tan B + \tan C = 3\sqrt{3} \quad \dots(1)$$

Now using A.M. $\geq$ G.M. (equality occurs when no's are equal)

For $\tan A, \tan B, \tan C$, we get

$$\frac{\tan A + \tan B + \tan C}{3} \geq (\tan A \tan B \tan C)^{1/3}$$

But in any ΔABC , we know that

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C$$

$\therefore$ Last inequality becomes

$$\frac{\tan A + \tan B + \tan C}{3} \geq (\tan A + \tan B + \tan C)^{1/3}$$

$$\Rightarrow (\tan A + \tan B + \tan C)^{2/3} \geq 3$$

$$\Rightarrow \tan A + \tan B + \tan C \geq 3\sqrt{3}$$

where equality occurs when $\tan A, \tan B, \tan C$ are equal, i.e., $A = B = C$

$$\Rightarrow \Delta ABC \text{ is equilateral.}$$

29. In ΔABC , O and I are circumcentre and incentre of Δ respectively and R and r are the respective radii of circum circle and incircle.

$$\text{To prove } (IO)^2 = R^2 - 2Rr$$

First of all we will find IO . Using cosine law in ΔAOI

$$\cos \angle OAI = \frac{OA^2 + AI^2 - OI^2}{2 \cdot OA \cdot AI} \quad \dots (1)$$

$$\text{where } OA = R$$

$$\text{In } \Delta AID, \sin A/2 = \frac{r}{AI}$$

$$AI = \frac{r}{\sin A/2}$$

$$AI = 4R \sin B/2 \sin C/2$$

[Using $r = 4R \sin A/2 \sin B/2 \sin C/2$]

$$\text{Also, } \angle OAI = \angle IAE - \angle OAE$$

$$= \frac{A}{2} - (90^\circ - \angle AOE)$$

$$= \frac{A}{2} - 90^\circ + \frac{1}{2} \angle AOC$$

$$= \frac{A}{2} - 90^\circ + \frac{1}{2} \cdot 2B \quad (\because O \text{ is circumcentre})$$

$$= \frac{A}{2} + B - \frac{A+B+C}{2}$$

$$= \frac{B-C}{2}$$

Substituting all these values in equation (1) we get

$$\cos \frac{B-C}{2} = \frac{R^2 + 16R^2 \sin^2 B/2 \sin^2 C/2 - OI^2}{2 \cdot R \cdot 4R \sin B/2 \sin C/2}$$

$$\Rightarrow OI^2 = R^2 + 16R^2 \sin^2 B/2 \sin^2 C/2$$

$$- 8R^2 \sin B/2 \sin C/2 \cos \frac{B-C}{2}$$

$$= R^2 \left[1 + \sin B/2 \sin C/2 \left\{ 2 \sin B/2 \sin C/2 - \cos \frac{B-C}{2} \right\} \right]$$

$$= R^2 \left[1 + 8 \sin B/2 \sin C/2 \left\{ 2 \sin B/2 \sin C/2 - \cos \frac{B}{2} \cos \frac{C}{2} - \sin B/2 \sin C/2 \right\} \right]$$

$$= R^2 \left[1 + 8 \sin B/2 \sin C/2 \left\{ \sin B/2 \sin C/2 - \cos B/2 \cos C/2 \right\} \right]$$

$$= R^2 \left[1 - 8 \sin B/2 \sin C/2 \cos \frac{B+C}{2} \right]$$

$$= R^2 \left[1 - 8 \sin A/2 \sin B/2 \sin C/2 \right]$$

$$= R^2 - 2R \cdot 4R \sin A/2 \sin B/2 \sin C/2$$

$$= R^2 - 2Rr.$$

Hence Proved.

Again if ΔOIB is right $\angle$ ed Δ then

$$\Rightarrow OB^2 = OI^2 + IB^2$$

$$\Rightarrow R^2 = R^2 - 2Rr + \frac{r^2}{\sin^2 B/2} \left[\because \text{In } \Delta IBD \sin B/2 = \frac{r}{IB} \right]$$

$$\Rightarrow 2R \sin^2 B/2 = r$$

$$\Rightarrow 2 \frac{abc}{4\Delta} \cdot \frac{(s-a)(s-c)}{ac} = \frac{\Delta}{s}$$

$$\Rightarrow b(s-a)(s-c) = 2(s-a)(s-b)(s-c)$$

$$\Rightarrow b = 2s - 2b \Rightarrow b = a + c - b$$

$$\Rightarrow a + c = 2b \Leftrightarrow a, b, c \text{ are in A.P.}$$

$$\Rightarrow b \text{ is A.M. between } a \text{ and } c. \text{ Hence Proved.}$$

30. Let $MN = r_3 = MP = MQ$, $ID = r$

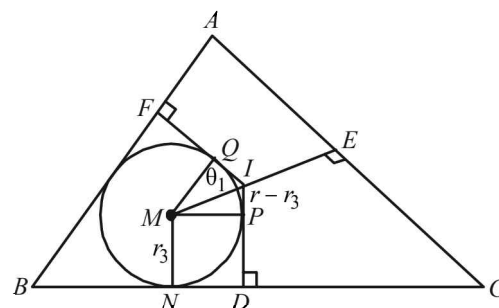
$$\Rightarrow IP = r - r_3$$

Clearly IP and IQ are tangents to circle with centre M .

$\therefore IM$ must be the $\angle$ bisector of $\angle PIQ$

$$\therefore \angle PIM = \angle QIM = \theta_1$$

$$\text{Also from } \Delta IPM, \tan \theta_1 = \frac{r_3}{r - r_3} = \frac{MP}{IP}$$



Here $DI = r$

Similarly, in other quadrilaterals, we get

$$\tan \theta_2 = \frac{r_2}{r - r_2} \text{ and } \tan \theta_3 = \frac{r_1}{r - r_1}$$

$$\text{Also } 2\theta_1 + 2\theta_2 + 2\theta_3 = 2\pi \Rightarrow \theta_1 + \theta_2 + \theta_3 = \pi$$

$$\Rightarrow \tan \theta_1 + \tan \theta_2 + \tan \theta_3 = \tan \theta_1 \cdot \tan \theta_2 \cdot \tan \theta_3$$

$$\Rightarrow \frac{r_1}{r - r_1} + \frac{r_2}{r - r_2} + \frac{r_3}{r - r_3} = \frac{r_1 r_2 r_3}{(r - r_1)(r - r_2)(r - r_3)}$$

31. We know, $\Delta = \sqrt{s(s-a)(s-b)(s-c)}$

$$= \sqrt{\frac{s}{8}(b+c-a)(c+a-b)(a+b-c)}$$

Since sum of two sides is always greater than third side;

$$\therefore b+c-a, c+a-b, a+b-c > 0$$

$$\Rightarrow (s-a)(s-b)(s-c) > 0$$

$$\text{Let } s-a = x, s-b = y, s-c = z$$

$$\text{Now, } x+y = 2s-a-b = c$$

$$\text{Similarly, } y+z = a \text{ and } z+x = b$$

Since $AM \geq GM$

$$\Rightarrow \frac{x+y}{2} \geq \sqrt{xy} \Rightarrow 2\sqrt{xy} \leq a \quad \frac{y+z}{2} \geq \sqrt{yz} \Rightarrow 2\sqrt{yz} \leq b$$

$$\frac{z+x}{2} \geq \sqrt{zx} \Rightarrow 2\sqrt{zx} \leq c \therefore 8xyz \leq abc$$

$$\Rightarrow (s-a)(s-b)(s-c) \leq \frac{1}{8} abc$$

$$\Rightarrow s(s-a)(s-b)(s-c) \leq \frac{sabc}{8}$$

$$\Rightarrow \leq \frac{1}{16} (a+b+c) abc \Rightarrow \Delta \leq \frac{1}{4} \sqrt{abc(a+b+c)}$$

and equality holds when $x = y = z \Rightarrow a = b = c$

32. Let OAB be one triangle out of n of a n sided polygon inscribed in a circle of radius 1.

Then $\angle AOB = \frac{2\pi}{n}$

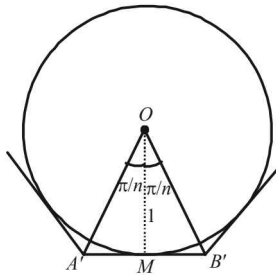
$$OA = OB = 1$$

$\therefore$ Using Area of isosceles Δ with vertical $\angle \theta$ and equal sides as

$$r = \frac{1}{2} r^2 \sin \theta = \frac{1}{2} \sin \frac{2\pi}{n}$$

$$\therefore I_n = \frac{n}{2} \sin \frac{2\pi}{n} \quad \dots (1)$$

Further consider the n sided polygon subscribing on the circle.



$A'MB'$ is the tangent of the circle at M .

$$\Rightarrow A'MB' \perp OM$$

$\Rightarrow A'MO$ is right angled triangle, right angle at M .

$$A'M = \tan \frac{\pi}{n}$$

$$\text{Area of } \Delta A'MO = \frac{1}{2} \times 1 \times \tan \frac{\pi}{n}$$

$$\therefore \text{Area of } \Delta A'B'O = \tan \frac{\pi}{n}$$

$$\text{So, } O_n = n \tan \frac{\pi}{n} \quad \dots (2)$$

Now, we have to prove

$$I_n = \frac{O_n}{2} \left(1 + \sqrt{1 - \left(\frac{2I_n}{n} \right)^2} \right) \text{ or } \frac{2I_n}{O_n} - 1 = \sqrt{1 - \left(\frac{2I_n}{n} \right)^2}$$

$$\text{LHS} = \frac{2I_n}{O_n} - 1 = \frac{n \sin \frac{2\pi}{n}}{n \tan \frac{\pi}{n}} - 1 \quad (\text{From (1) and (2)})$$

$$= 2 \cos^2 \frac{\pi}{n} - 1 = \cos \frac{2\pi}{n}$$

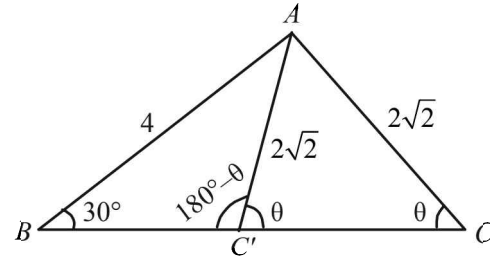
$$\text{RHS} = \sqrt{1 - \left(\frac{2I_n}{n} \right)^2} = \sqrt{1 - \sin^2 \frac{2\pi}{n}} \quad (\text{From (1)})$$

$$= \cos \left(\frac{2\pi}{n} \right)$$

Hence Proved.

I. Integer Value Correct Type

1. (4) Let $\angle ACC' = \theta$, then $\angle AC'C = \theta$ ($\because AC = AC'$) and $\angle AC'B = 180 - \theta$.



Applying sine law in $\Delta ABC'$, we get

$$\frac{4}{\sin(180 - \theta)} = \frac{2\sqrt{2}}{\sin 30^\circ} \Rightarrow \sin \theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$$

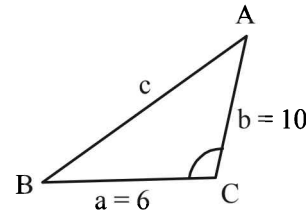
$$\therefore \angle CAC' = 90^\circ$$

So, the required area = $ar(\Delta ABC) - ar(\Delta ABC')$

$$= ar(\Delta ACC') = \frac{1}{2} \times AC \times AC'$$

$$= \frac{1}{2} \times 2\sqrt{2} \times 2\sqrt{2} = 4 \text{ sq. units.}$$

2. (3)



We know that area of $\Delta = \frac{1}{2} ab \sin C$

$$\Rightarrow \frac{1}{2} \times 6 \times 10 \times \sin C = 15\sqrt{3}$$

$$\Rightarrow \sin C = \frac{\sqrt{3}}{2}$$

$\Rightarrow C = 120^\circ$ as C is obtuse angle.

$$\text{Now using } \cos C = \frac{a^2 + b^2 - c^2}{2ab},$$

$$\text{we get } \cos 120^\circ = \frac{36 + 100 - c^2}{120}$$

$$\Rightarrow c^2 = 196 \text{ or } c = 14$$

$$\therefore s = \frac{a+b+c}{2} = 15$$

$$\text{There radius of incircle, } r = \frac{\Delta}{s} = \frac{15\sqrt{3}}{15} = \sqrt{3}$$

$$\therefore r^2 = 3$$

Section-B

JEE Main/ AIEEE

1. (b) Let $a = 3x + 4y$, $b = 4x + 3y$ and $c = 5x + 5y$
as $x, y > 0$, $c = 5x + 5y$ is the largest side
 $\therefore C$ is the largest angle. Now

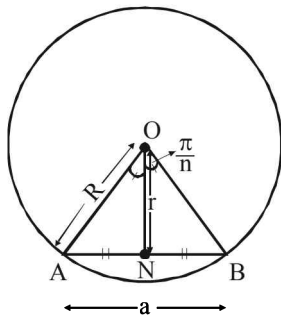
$$\cos C = \frac{(3x+4y)^2 + (4x+3y)^2 - (5x+5y)^2}{2(3x+4y)(4x+3y)}$$

$$= \frac{-2xy}{2(3x+4y)(4x+3y)} < 0$$

$\therefore C$ is obtuse angle $\Rightarrow \Delta ABC$ is obtuse angled

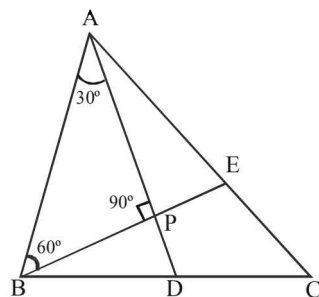
2. (a) $r_1 > r_2 > r_3 \Rightarrow \frac{\Delta}{s-a} > \frac{\Delta}{s-b} > \frac{\Delta}{s-c}$
 $\Rightarrow s-a < s-b < s-c \Rightarrow -a < -b < -c \Rightarrow a > b > c$

3. (c) $\tan\left(\frac{\pi}{n}\right) = \frac{a}{2r}$; $\sin\left(\frac{\pi}{n}\right) = \frac{a}{2R}$
 $r + R = \frac{a}{2} \left[\cot \frac{\pi}{n} + \csc \frac{\pi}{n} \right]$



$$= \frac{a}{2} \left[\frac{\cos \frac{\pi}{n} + 1}{\sin \frac{\pi}{n}} \right] = \frac{a}{2} \left[\frac{2 \cos^2 \frac{\pi}{2n}}{2 \sin \frac{\pi}{2n} \cos \frac{\pi}{2n}} \right] = \frac{a}{2} \cot \frac{\pi}{2n}$$

4. (d)



$$AP = \frac{2}{3} AD = \frac{8}{3}; \quad PD = \frac{4}{3}; \quad \text{Let } PB = x$$

$$\tan 60^\circ = \frac{8/3}{x} \text{ or } x = \frac{8}{3\sqrt{3}}$$

$$\text{Area of } \Delta ABD = \frac{1}{2} \times 4 \times \frac{8}{3\sqrt{3}} = \frac{16}{3\sqrt{3}}$$

$$\therefore \text{Area of } \Delta ABC = 2 \times \frac{16}{3\sqrt{3}} = \frac{32}{3\sqrt{3}}$$

[$\because$ Median of a Δ divides it into two Δ 's of equal area.]

5. (b) If $a \cos^2\left(\frac{C}{2}\right) + c \cos^2\left(\frac{A}{2}\right) = \frac{3b}{2}$

$$a[\cos C + 1] + c[\cos A + 1] = 3b$$

$$(a+c) + (a \cos C + c \cos A) = 3b$$

$$a+c+b = 3b \text{ or } a+c = 2b \text{ or } a, b, c \text{ are in A.P.}$$

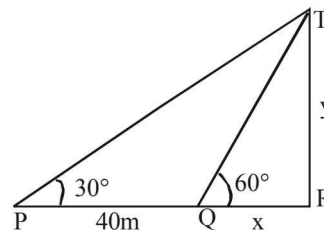
6. (c) Let $a = \sin \alpha$, $b = \cos \alpha$ and $c = \sqrt{1 + \sin \alpha \cos \alpha}$
Clearly a and $b < 1$ but $c > 1$ as $\sin \alpha > 0$ and $\cos \alpha > 0$
 $\therefore c$ is the greatest side and greatest angle is C

$$\therefore \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$= \frac{\sin^2 \alpha + \cos^2 \alpha - 1 - \sin \alpha \cos \alpha}{2 \sin \alpha \cos \alpha} = -\frac{1}{2}$$

$$\therefore C = 120^\circ$$

7. (d)



From the figure

$$\tan 60^\circ = \frac{y}{x} \Rightarrow y = \sqrt{3}x \dots\dots\dots(1)$$

$$\tan 30^\circ = \frac{y}{x+40} \Rightarrow y = \frac{x+40}{\sqrt{3}} \dots\dots\dots(2)$$

$$\text{From (1) and (2), } \sqrt{3}x = \frac{x+40}{\sqrt{3}} \Rightarrow x = 20m$$

8. (b) We know by sine rule $\frac{c}{\sin C} = 2R \Rightarrow c = 2R \sin C$

$$\Rightarrow c = 2R \quad (\because \angle C = 90^\circ)$$

$$\text{Also } \tan \frac{C}{2} = \frac{r}{s-c}$$

$$\Rightarrow \tan \frac{\pi}{4} = \frac{r}{s-c} \quad (\because \angle C = 90^\circ)$$

$$\Rightarrow r = s - c = \frac{a+b-c}{2}$$

$$\Rightarrow 2r + c = a + b \Rightarrow 2r + 2R = a + b \quad (\text{using } c = 2R)$$

9. (b) $\Delta = \frac{1}{2} p_1 a = \frac{1}{2} p_2 b = \frac{1}{2} p_3 c$

p_1, p_2, p_3 , are in H.P.

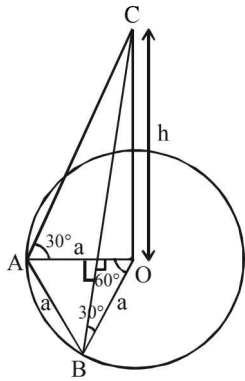
$$\Rightarrow \frac{2\Delta}{a}, \frac{2\Delta}{b}, \frac{2\Delta}{c} \text{ are in H.P.} \Rightarrow \frac{1}{a}, \frac{1}{b}, \frac{1}{c} \text{ are in H.P.}$$

$$\Rightarrow a, b, c \text{ are in A.P.}$$

$$\Rightarrow K \sin A, K \sin B, K \sin C \text{ are in AP}$$

$$\Rightarrow \sin A, \sin B, \sin C \text{ are in A.P.}$$

10. (a) In the $\triangle AOB$, $\angle AOB = 60^\circ$, and $\angle OBA = \angle OAB$ (since $OA = OB = AB$ radius of same circle). $\therefore \triangle AOB$ is an equilateral triangle. Let the height of tower is h



m . Given distance between two points A & B lie on boundary of circular park, subtends an angle of 60° at the foot of the tower is AB i.e. $AB = a$. A tower OC stands at the centre of a circular park. Angle of elevation of the top of the tower from A and B is 30° .

$$\text{In } \triangle OAC \tan 30^\circ = \frac{h}{a} \Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{a} \Rightarrow h = \frac{a}{\sqrt{3}}$$

11. (b) In $\triangle ABC$

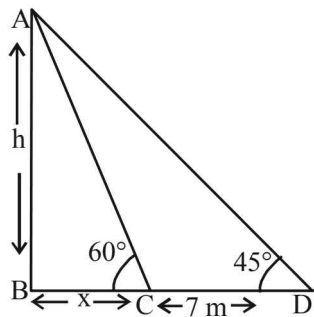
$$\frac{h}{x} = \tan 60^\circ = \sqrt{3}$$

$$\Rightarrow x = \frac{h}{\sqrt{3}}$$

$$\text{In } \triangle ABD \frac{h}{x+7} = \tan 45^\circ = 1$$

$$\Rightarrow h = x + 7 \Rightarrow h - \frac{h}{\sqrt{3}} = 7$$

$$\Rightarrow h = \frac{7\sqrt{3}}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} \Rightarrow h = \frac{7\sqrt{3}}{2} (\sqrt{3}+1) m$$

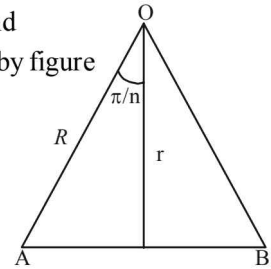


12. (b) If O is centre of polygon and AB is one of the side, then by figure

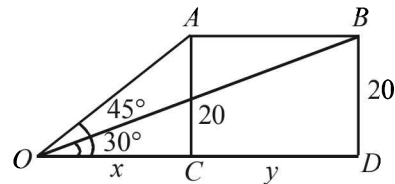
$$\cos \frac{\pi}{n} = \frac{r}{R}$$

$$\Rightarrow \frac{r}{R} = \frac{1}{2}, \frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{2} \text{ for}$$

$$n = 3, 4, 6 \text{ respectively.}$$



13. (b) Let the speed be y m/sec.
Let AC be the vertical pole of height 20 m.
Let O be the point on the ground such that $\angle AOC = 45^\circ$
Let $OC = x$



Time $t = 1$ s

$$\text{From } \triangle AOC, \tan 45^\circ = \frac{20}{x} \quad \dots (i)$$

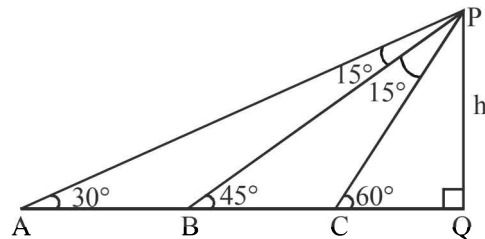
$$\text{and from } \triangle BOD, \tan 30^\circ = \frac{20}{x+y} \quad \dots (ii)$$

$$\text{From (i) and (ii), we have } x = 20 \text{ and } \frac{1}{\sqrt{3}} = \frac{20}{x+y}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{20}{20+y} \Rightarrow 20+y = 20\sqrt{3}$$

$$\text{So, } y = 20(\sqrt{3}-1) \text{ i.e., speed} = 20(\sqrt{3}-1) \text{ m/s}$$

14. (c)



$\therefore$ PB bisects $\angle APC$, therefore $AB : BC = PA : PC$

$$\text{Also in } \triangle APQ, \sin 30^\circ = \frac{h}{PA} \Rightarrow PA = 2h$$

$$\text{and in } \triangle CPQ, \sin 60^\circ = \frac{h}{PC} \Rightarrow PC = \frac{2h}{\sqrt{3}}$$

$$\therefore AB : BC = 2h : \frac{2h}{\sqrt{3}} = \sqrt{3} : 1$$